2025 RATIONALIZED Grade 9 Rationalized Mathematics Lesson Plans Term 2

WEEK 1: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, students should be able to:
- Identify the properties of regular polygons.
- Calculate the sum of the interior and exterior angles of various polygons.
- Acknowledge the properties of different polygons.

Key Inquiry Question:
- How do you find the sum of interior and exterior angles of regular polygons?

Learning Resources:
- Top Scholar Maths, pg 87
- Lesson notes
- Charts

Organisation of Learning:

Introduction (5 minutes)
1. Review the previous lesson focused on shapes and their characteristics.
2. Facilitate a discussion where students share what they remember about angles and polygons. Encourage them to read key sections from their textbook relevant to today's topic.

Lesson Development (30 minutes)

Step 1: Identify Polygons (10 minutes)
- In small groups, students will discuss and identify different types of regular polygons (e.g., triangle, square, hexagon, pentagon).
- Each group will list the common characteristics of these polygons (equal sides, equal angles).

Step 2: Drawing Polygons (10 minutes)

- Students will use charts and notebooks to draw an equilateral triangle, a square, a regular pentagon, and a regular hexagon.
- Ensure they label the number of sides and angles for each shape.

Step 3: Discussing Properties (5 minutes)
- Groups will discuss the properties of the polygons they have drawn, focusing on side lengths, angles, and symmetry. Encourage students to share these properties with the class.

Step 4: Calculating Angles (5 minutes)
- Introduce the formulas for finding the sum of the interior angles and the exterior angles
- Each group will apply these formulas to calculate the sum of interior and exterior angles for the polygons they drew and share their calculations with the class.

Conclusion (5 minutes)
- Summarize the key points covered: types of polygons, their properties, and angle calculations.
- Conduct a brief interactive quiz where students answer questions based on the properties and angle sums discussed.
- Provide a preview of the next lesson, focusing on the relationship between the properties of polygons and their areas.

Extended Activities:
- Have students research real-life applications of regular polygons (e.g., architecture, art) and present their findings in the next class.
- Create a project where students find various regular polygon patterns in their environment and take photographs to present to the class with calculated angle sums.

Teacher Self-Evaluation:

WEEK 1: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify the formula for calculating the area of a pentagon.
- Calculate the area of a pentagon in different situations.
- Acknowledge the formula for calculating the area of a pentagon.

Key Inquiry Question:
- How do we work out the area of a pentagon?

Learning Resources:
- Cut outs of pentagons
- Top Scholar Maths pg 87-89
- Lesson notes
- Teacher's Guide
- Rulers and manillas

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of the key concepts related to area measurement.

Lesson Development (30 minutes)

Step 1: Introduction to the Pentagon
- Show a cut-out of a pentagon.
- Discuss the properties of a pentagon (5 sides, etc.).
- Guide students to brainstorm what they already know about calculating area.

Step 2: Formula Identification
- Engage learners in groups to explore resources and derive the formula for the area of a pentagon.
- Facilitate a discussion to ensure they understand the structure of the formula:

$$ A = \frac{1}{2} \times Perimeter \times Apothem $$

Step 3: Demonstration
- Illustrate the process of calculating the area using cut-outs of pentagons.
- Explain how to measure the sides and the apothem using rulers.
- Choose an example with specific measurements and work through the calculation as a class.

Step 4: Group Calculation Activity
- Provide each group with different cut-outs and measurements for pentagons.
- Instruct them to calculate the area of their pentagons.
- Encourage groups to share their findings with the class, discussing any differences in results and what may have caused them.

Conclusion (5 minutes)
- Summarize key points: the formula for the area of a pentagon and the steps to use it.
- Conduct a brief interactive activity, such as a quiz or a reflection exercise in pairs, reinforcing the main topics discussed.
- Prepare learners for the next session by previewing upcoming topics, like the area of other polygons.

Extended Activities:
- Outdoor Activity: Measure and calculate the area of pentagonal-shaped objects in the schoolyard or home.
- Research Project: Investigate how the area of pentagons is used in real-world applications, such as architecture or art.
- Creative Math: Create a colorful pentagonal artwork and calculate the area based on the dimensions used in their design.

Teacher Self-Evaluation:

WEEK 1: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify the formula for calculating the area of a hexagon.
- Calculate the area of hexagons in different situations.
- Enjoy working out the area of hexagons.

Key Inquiry Question:
- How do you find the area of hexagons?

Learning Resources:
- Top Scholar Maths pg 90-91
- Cut-outs of hexagons
- Teacher's guide
- Calculators

Organisation of Learning:

Introduction (5 minutes)
1. Review the previous lesson, focusing on polygon areas.
2. Guide learners to read the relevant sections from the learning resource on hexagons. Facilitate a discussion that emphasizes understanding the unique properties of hexagons and introduces the key inquiry question.

Lesson Development (30 minutes)

Step 1: Prepare Cut-outs
- In pairs or small groups, learners will use the provided materials to prepare cut-outs of regular hexagons.
- Each group should measure the side lengths to ensure accuracy.

Step 2: Identify the Formula
- Learners will brainstorm or recall what they know about finding the area of polygons.
- The teacher will guide them to discover the formula for the area of a hexagon,

Step 3: Teacher Demonstration
- The teacher will demonstrate how to calculate the area using one of the cut-outs while guiding students through each step of the calculation.
- Emphasize the substitution of the side length into the formula.

Step 4: Collaborative Calculation
- Learners will work in their groups to calculate the area of their hexagon cut-outs.
- They will share their calculations with the class, explaining their process and answers.

Conclusion (5 minutes)
- Summarize the key points regarding the formula for the area of a hexagon and the calculation process.
- Conduct a brief interactive activity, such as a quiz or a quick round of "Hexagon Fact or Fiction" to reinforce the main topics.
- Prepare learners for the next session by previewing the topic of calculating areas of irregular shapes or how hexagons are used in real-life scenarios.

Extended Activities:
- Design a project where students research real-life applications of hexagons in nature and architecture and present their findings.
- Create a worksheet with various hexagon dimensions and scenarios where students can practice calculating the area.
- Challenge students to create their own hexagon-shaped designs, calculate their areas, and explain their design’s significance.

Teacher Self-Evaluation:

WEEK 1: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Outline the steps of making models of triangular and rectangular based prisms using manillas.
- Make models of triangular and rectangular based prisms using manillas.
- Enjoy making triangular and rectangular based prisms using manillas.

Key Inquiry Question:
How can you make models of prisms?

Learning Resources:
- Manillas
- Scissors
- Pencils and rulers
- Top Scholar Maths pg 92
- Digital devices
- Video clips

Organisation of Learning:

Introduction (5 minutes)
1. Briefly review the previous lesson, focusing on any relevant concepts related to shapes and measurements.
2. Guide learners to read and discuss the relevant content from the learning resources, emphasizing key concepts such as the properties of triangular and rectangular prisms.

Lesson Development (30 minutes)

Step 1: Research and Exploration (10 minutes)
- In groups, learners use digital devices to find and watch video clips demonstrating how to make models of triangular and rectangular based prisms using manillas.

Step 2: Sketching the Nets (10 minutes)

- Learners sketch the nets of the triangular and rectangular prisms based on the videos they watched. They should ensure that their sketches are labeled and include all necessary dimensions.

Step 3: Model Making (10 minutes)
- Guided by the teacher, groups collaborate to cut manillas and assemble their models of both triangular and rectangular based prisms. Encourage creativity in their models and remind them to work together.

Step 4: Display and Share (5 minutes)
- Each group presents their models to the class, explaining the shapes and dimensions of their prisms. Encourage classmates to ask questions or provide feedback on the models.

Conclusion (5 minutes)
- Summarize the key points from the lesson: understanding the properties of prisms and the practical process of creating models.
- Conduct a brief interactive activity, such as passing around a prism model and having students mention a property or formula related to prisms.
- Prepare learners for the next session by previewing upcoming topics, such as surface area and volume of prisms, or posing the question: "Why are prisms important in real life?"

Extended Activities:
- Have students create a mini-project in which they research and present on different types of prisms found in architecture or nature.
- Assign a homework task to create a 3D model of a prism using recyclable materials, encouraging them to calculate the surface area and volume of their creation.
- Invite students to explore real-world applications of prisms in fields such as engineering or art, and write a short reflection on what they learned.

Teacher Self-Evaluation:

WEEK 1: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify the steps for calculating the surface area of a rectangular based prism.
- Work out the surface area of a rectangular based prism.
- Enjoy working out surface area of rectangular prisms.

Key Inquiry Question(s):
- How do you calculate the surface area of rectangular based prisms?

Learning Resources:
- Drawn nets of rectangular prisms
- Top Scholar Maths (pg 92-93)
- Teacher's Guide

Organisation of Learning:

Introduction (5 minutes)
- Review: Begin with a quick review of the previous lesson, discussing the properties of three-dimensional shapes.
- Discussion: Guide learners to read and discuss relevant content from the learning resources provided, emphasizing understanding of surface area.

Lesson Development (30 minutes)

Step 1: Understanding Nets (10 minutes)
- Instruct students to observe the drawn nets of rectangular prisms.
- Discuss what a net is (a two-dimensional representation of a three-dimensional object).
- Allow them to identify the different faces (length, width, height) of the prism.

Step 2: Formulating the Surface Area Formula (10 minutes)
- Guide learners to list the steps for calculating the surface area:
1. Identify all the dimensions (length, width, height).
2. Use the formula: Surface Area (SA) = 2(lb + lh + wh).
3. Discuss how each area of the faces contributes to the total surface area.

Step 3: Group Activity - Drawing and Calculating (5 minutes)
- In groups, learners draw their own nets of a rectangular prism in their books.
- Calculate the surface area using dimensions provided (e.g., length = 5 cm, width = 3 cm, height = 4 cm).

Step 4: Sharing and Feedback (5 minutes)
- Groups share their findings with the class.
- Encourage peer assessment, where classmates provide constructive feedback on calculations and presentation of nets.

Conclusion (5 minutes)
- Summarize: Highlight the key points about surface area and the steps to calculate it.
- Interactive Activity: Conduct a quick quiz where students must identify which part of the net corresponds to which formula component.
- Preview: Briefly introduce the next session topic on volume, encouraging students to think about the relationship between area and volume.

Extended Activities:
- Net Creation Challenge: Have students create different nets for various dimensions of rectangular prisms and calculate their surface areas.
- Real-Life Applications: Research real-life objects that are rectangular prisms (e.g., boxes, buildings) and calculate their surface area using measurements.

Teacher Self-Evaluation:

WEEK 2: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Outline the steps of calculating the area of triangular-based prisms.
- Work out the surface area of triangular-based prisms.
- Enjoy working out the surface area of triangular-based prisms.

Key Inquiry Question:
- How do we calculate the surface area of triangular prisms?

Learning Resources:
- Top Scholar Maths, pg 93-95
- Models and drawn nets of triangular prisms
- Charts
- Teacher's Guide
- Chalkboard & chalks

Organisation of Learning

Introduction (5 minutes)
- Review: Begin by briefly reviewing the previous lesson on basic area calculations.
- Discussion: Guide learners to read and discuss the relevant content from the learning resources. Highlight the importance of understanding triangular prisms and their surfaces.

Lesson Development (30 minutes)

Step 1: Understanding Nets of Triangular Prisms (10 minutes)
- Activity: Each student draws a net of a triangular-based prism on paper.
- Guidance: Explain what a net is and how it represents the 3D shape in 2D.
- Discussion: Discuss the different parts of the prism and what shapes are found in the net.

Step 2: Calculating Surface Area (10 minutes)
- Instruction: Present a formula used to calculate the surface area of triangular prisms.
- Guided Practice: Walk through an example with the class explaining each step: finding the area of the triangular base, calculating the area of the rectangular faces, and summing them together.

Step 3: Group Work (10 minutes)
- Activity: In small groups, learners will use the drawn nets and models to calculate the surface area of different triangular-based prisms provided by the teacher.
- Collaboration: Encourage students to discuss their approaches within their groups to develop understanding and correct mutual errors.

Step 4: Sharing Findings (Optional if time permits)
- Presentation: Groups share their calculations and reasoning with the class. Provide constructive feedback and prompt questions to reinforce understanding and clarify uncertainties.

Conclusion (5 minutes)
- Summarize: Recap the key points discussed in the lesson, emphasizing the steps to calculate surface area.
- Interactive Activity: Conduct a brief quiz or a fun game (e.g., Kahoot!) where students answer questions related to surface area calculations.
- Preview: Prepare learners for the next session by posing questions to consider about volume calculations related to prisms.

Extended Activities
- Homework: Assign students to find various real-life objects shaped like triangular prisms (like tents or kites) and calculate their surface areas as a hands-on project.
- Creative Project: Have students create their own triangular-based prism shape and calculate the surface area, presenting their findings in class.
- Technology Integration: Explore online geometry tools that allow the construction and manipulation of triangular prisms.

Teacher Self-Evaluation:

WEEK 2: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
- Identify the steps of calculating the surface area of triangular based pyramids.
- Work out the surface area of triangular based pyramids.
- Acknowledge the steps of working out the surface area of triangular based pyramids.

Key Inquiry Question(s):
- How do you calculate the surface area of triangular based pyramids?

Learning Resources:
- Top Scholar Maths pg 97-98
- Video clip on calculating surface area of triangular pyramids
- Digital devices
- Nets of triangular based pyramids
- Teacher's Guide
- Chalkboard & chalks

Organisation of Learning

Introduction (5 minutes)
- Review the content from the previous lesson on surface area and volume of solids. Engage students in a brief discussion to recall key terms and formulas.
- Guide learners to read relevant sections from the Top Scholar Maths resources, focusing on the surface area of triangular based pyramids.

Lesson Development (30 minutes)

Step 1: Understanding the Concept (10 minutes)
- Introduce the definition of triangular based pyramids. Discuss its properties briefly and explain the significance of calculating surface area.
- Show the nets of triangular based pyramids on the board. Ask students to observe and describe the different triangular faces.

Step 2: Formula Identification (5 minutes)

- Encourage students to explore the video clip showing how to calculate the surface area of triangular based pyramids.
- In groups, have students identify and write down the formula for the surface area based on the video and their own notes.

Step 3: Application of Formula (10 minutes)
- Distribute nets of triangular based pyramids to each student or group.
- Have students calculate the surface area using the formula they identified in the previous step. Monitor them as they apply the calculations and provide assistance where necessary.

Step 4: Sharing and Feedback (5 minutes)
- Allow students to share their findings with their peers. Have groups present their calculations and reasoning to the class.
- Encourage constructive feedback amongst peers on the methods used in calculating the surface area.

Conclusion (5 minutes)
- Summarize the key points discussed: definition, properties, formula, and calculations related to the surface area of triangular based pyramids.
- Conduct a quick interactive quiz where students can shout out answers to reinforce the formulas and concepts learned.
- Preview the next lesson by mentioning upcoming topics, such as the surface area and volume comparison of different shapes and their real-world applications.

Extended Activities
- Building Models: Have students create three-dimensional models of triangular based pyramids using paper or cardboard, then calculate the surface area of their models.
- Real-Life Application: Research and present on objects in real life that are triangular based pyramids, such as certain tents, roofs, or sculptures, and analyze their surface areas.
- Interactive Game: Create an online Kahoot! quiz that covers the concepts learned during the lesson, helping reinforce knowledge in a fun and engaging way.

Teacher Self-Evaluation:

WEEK 2: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, the learner will be able to:
- Identify the steps and formula for calculating the surface area of square-based pyramids.
- Work out the surface area of square-based pyramids using nets.
- Enjoy working out the surface area of square-based pyramids.

Key Inquiry Question(s):
- How do we find the surface area of square-based pyramids?

Learning Resources:
- Top Scholar Maths pg 95-96 & 98
- Video clips
- Digital devices
- Teacher's Guide
- Nets of square-based pyramids
- Chalkboard & chalks

Organisation of Learning

Introduction (5 minutes):
- Begin the lesson by reviewing the previous topic on area calculations.
- Ask students to recall what they learned about different shapes and their properties.
- Introduce today's topic by guiding them to read relevant sections from the learning resources and discuss the key concepts related to square-based pyramids.

Lesson Development (30 minutes):

Step 1: Understanding the Pyramid
- Show a square-based pyramid and discuss its components (base, height, slant height, faces).
- Explain the concept of a net and how it represents the flattened shape of the pyramid.
- Have students examine the nets of square-based pyramids provided in class.

Step 2: Deriving the Formula
- Guide learners through the process of calculating the surface area of a square-based pyramid.

- Present the formula: Surface Area = Base Area + Lateral Area.
- Explain that the base is a square ($A = side^2$) and the lateral area can be calculated using triangles (Area = 1/2 * base * height).

Step 3: Group Activity - Drawing Nets
- In groups, assign students to draw their own nets of a square-based pyramid on chart paper.
- Each group should label the parts (base, triangular faces) and dimensions.

Step 4: Calculating Surface Area
- Using the drawn nets, have students practice calculating the surface area of different square-based pyramids.
- Ask them to use the identified formulas and share their calculations with their peers for feedback.

Conclusion (5 minutes):
- Summarize the key points discussed, including the formula and significance of calculating surface area.
- Conduct a quick interactive quiz to reinforce understanding (e.g., ask questions like "What is the formula for the base area?").
- Preview the next lesson, which will explore volume and how it relates to surface area in three-dimensional shapes.

Extended Activities:
- Encourage students to find real-life examples of square-based pyramids (e.g., tents, certain buildings) and calculate their surface areas using the dimensions they can gather.
- Assign a project where students create a 3D model of a square-based pyramid and calculate its surface area.

Teacher Self-Evaluation:

WEEK 2: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify the steps and formula for calculating the surface area of rectangular based pyramids.
- Work out the surface area of rectangular based pyramids.
- Enjoy working out the surface area of rectangular based pyramids using their nets.

Key Inquiry Question(s):
- How do we calculate the surface area of rectangular based pyramids?

Learning Resources:
- Top Scholar Maths pg 97-98
- Video clip on surface area calculations
- Digital devices for accessing online content
- Drawn nets of rectangular based pyramids
- Teacher's Guide

Organisation of Learning:

Introduction (5 minutes):
- Review the previous lesson focusing on the area of basic shapes.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of pyramids.

Lesson Development (30 minutes):

Step 1: Understanding Nets
- Explain what a net is and its importance in finding the surface area.
- Ask students to individually draw the net of a rectangular based pyramid in their notebooks or on chart paper.

Step 2: Video Demonstration
- In groups, learners will watch a video clip on finding the surface area of rectangular based pyramids.
- Prompt them to take notes on the steps and formula presented in the video.

Step 3: Discussion of Steps
- Reassemble as a class and discuss the steps observed in the video.
- Write down the formula for the surface area of a rectangular based pyramid (Surface Area = Base Area + Lateral Area) on the board and explain each part clearly.

Step 4: Applying Knowledge
- Using the nets they drew and the formula discussed, guide students as they calculate the surface area of given rectangular based pyramids on the board.
- Finally, encourage them to share their calculations and findings with their peers, fostering collaborative assessment and feedback.

Conclusion (5 minutes):
- Summarize the key points discussed, such as the importance of nets and the formula for surface area.
- Conduct a brief interactive activity, such as a quiz or a fun challenge where students identify the surface area of random pyramids based on prompts.
- Preview upcoming topics, such as the surface area of other shapes, and pose questions for them to ponder in the next session.

Extended Activities:
- Art Integration: Have students create a 3D model of their pyramid using card or paper, then calculate the surface area and compare it with their previous calculations.
- Research Activity: Assign students to research real-world applications of pyramids and present findings to the class, emphasizing their surface areas.
- Math Journals: Encourage learners to maintain a math journal to reflect on their learning, include diagrams of nets, and practice additional problems on their own.

Teacher Self-Evaluation:

WEEK 2: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Identify the formula for calculating the area of a sector and segment of a circle.
- Calculate the area of a sector and segment of a circle.
- Acknowledge the formulae for calculating the area of a sector and segment of a circle.

Key Inquiry Question(s):
- How do we find the area of a sector and segment of a circle?

Learning Resources:
- Top Scholar Maths, pages 98-100
- Teacher's Guide
- Chalkboard & chalks
- Chart showing the sector and segment

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of the key concepts.

Lesson Development (30 minutes)

Step 1: Introduction to Circle Components (10 minutes)
- Begin by explaining the parts of a circle: radius, diameter, sector, and segment.
- Use the chart to visually demonstrate a sector and a segment.
- Have students engage by asking questions about their observations.

Step 2: Group Activity - Drawing and Identifying (10 minutes)
- In groups, learners will use manila papers to draw a circle, marking a sector, chord, and segment.
- Each group discusses the relationships between these parts and identifies where they fit in the circle.

- Encourage groups to label their drawings.

Step 3: Formula Identification (5 minutes)
- Have each group share their drawings with the class.
- Write down the sector and segment formulas on the chalkboard:
- Area of a Sector = $(\theta/360) \times \pi r^2$
- Area of a Segment = Area of the sector - Area of the triangle formed by the radius and the chord.
- Guide students to identify each part of the formulas and what they represent.

Step 4: Practice Calculation (5 minutes)
- Provide examples of sectors and segments for groups to calculate areas using the formulas.
- Encourage collaboration in discussing the steps for each calculation.
- Ask groups to present their answers to reinforce peer learning.

Conclusion (5 minutes)
- Summarize key points and learning objectives achieved during the lesson: circle components, area formulas, and calculation practices.
- Conduct a brief interactive quiz or activity where students have to match parts of the circle to their corresponding area formulas.
- Preview what will be covered in the next session, encouraging learners to think about real-life applications of these calculations.

Extended Activities:
- Assign students to research real-world applications of sectors and segments, such as in architecture or astronomy, and present their findings in the next lesson.
- Encourage students to create a project that involves measuring and calculating areas of sectors and segments in various objects around them, like pizza slices or pie charts.

Teacher Self-Evaluation:

WEEK 3: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub-Strand: Area

Specific Learning Outcomes:
By the end of the lesson, students should be able to:
- Identify the formula for calculating the surface area of a cone.
- Work out the surface area of a cone in real-life situations.
- Enjoy calculating the surface area of cones in real-life situations.

Key Inquiry Question:
- How do we find the surface area of cones?

Learning Resources:
- Top Scholar Maths pg 101-102
- Nets of cones
- Manillas
- Teacher's Guide
- Video clip
- Digital devices

Organisation of Learning

Introduction (5 minutes)
- Begin with a review of the previous lesson on areas.
- Engage the class in a discussion about cones, prompting students to share what they know.
- Direct students to read and summarize the relevant sections from the learning resources, paying special attention to the concept of surface area.

Lesson Development (30 minutes)

Step 1: Understanding the Cone
- Have students brainstorm different objects that are shaped like cones (ice cream cones, funnels, etc.).
- Introduce the key parts of a cone: base radius (r) and height (h).

Step 2: Drawing the Net of a Cone
- In groups, students will create a net of a cone using manilla paper.

- Encourage each group to work together to label the parts (base, slant height) accurately.

Step 3: Exploring Surface Area through Digital Media
- Guide students to use digital devices to watch a video clip about calculating the surface area of a cone.
- After watching, ask groups to discuss and write down the formula for the surface area of a cone, emphasizing the curved surface area and the base:
$$\text{Surface Area} = \pi r (r + l)$$
Where l is the slant height.

Step 4: Real-Life Application
- Challenge groups with a real-life problem: Calculate the surface area of a cone-shaped object (e.g., a painted traffic cone). Provide the necessary dimensions.
- Each group presents their findings, explaining the steps they took to arrive at their answer.

Conclusion (5 minutes)
- Summarize the key points discussed about the surface area of cones.
- Engage students in a quick interactive quiz or game to reinforce their understanding of the material covered − perhaps a Kahoot! quiz on the surface area formula or properties of cones.
- Provide a preview of the next lesson, which will focus on volume calculations of cone-shaped objects. Pose questions for consideration, such as, "How does finding volume differ from finding surface area?"

Extended Activities:
- Assign students to find cone-shaped objects at home and calculate their surface areas using the formula learned in class.
- Encourage students to create a poster that explains the concept of surface area, including examples and diagrams of real-life cones.
- Have students research the volume of a cone and how it relates to surface area, preparing for further discussions in the next lesson.

Teacher Self-Evaluation:

WEEK 3: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of this lesson, learners should be able to:
- Identify the formula for calculating the surface area of a sphere.
- Calculate the surface area of a sphere in real-life situations.
- Enjoy calculating the surface area of spheres in different contexts.

Key Inquiry Question(s):
- How do you calculate the surface area of spheres?

Learning Resources:
- Top Scholar Mathematics pg 102-103
- Teacher's Guide
- Chalkboard & chalks

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on surface area.
- Engage learners in a discussion about familiar spherical objects (e.g., basketballs, oranges).
- Prompt learners to read and discuss relevant content from the provided learning resources.

Lesson Development (30 minutes)

Step 1: Identify the Formula (10 minutes)
- In pairs, guide learners to discover the formula for the surface area of a sphere: $SA = 4\pi r^2$, where SA is surface area and r is the radius.
- Have students discuss the meaning of terms in the formula and how they can identify the radius in real-life objects.

Step 2: Teacher Illustration (10 minutes)
- Use the chalkboard to illustrate how the formula is applied. Show an example using a specific sphere (e.g., a baseball) with a known radius.

- Walk through the calculation step-by-step, emphasizing the importance of using the correct measurement units.

Step 3: Hands-On Calculation (5 minutes)
- Provide learners with several different sizes of spheres (maybe images or tangible objects such as tennis balls, globes, etc.).
- Have them work in groups to measure the radius of their assigned spherical object and calculate its surface area using the identified formula.

Step 4: Sharing & Feedback (5 minutes)
- Invite each group to present their findings to the class.
- Encourage classmates to ask questions and provide constructive feedback on each presentation.

Conclusion (5 minutes)
- Summarize the key points: the formula for the surface area of a sphere and the method of calculation.
- Conduct a brief interactive quiz to reinforce concepts learned during the lesson.
- Prepare learners for the next session by introducing the topic of volume of spheres and posing the question: "How do we calculate the volume of a sphere?"

Extended Activities:
- Research Project: Students can choose a spherical object (e.g., various sports balls, globes) and calculate the surface area, presenting their findings through a poster or presentation.
- Real-World Application: Have students investigate how surface area affects the design of objects in real life, like sports equipment or packaging.
- Creative Task: Encourage students to design their own spherical object and calculate its surface area, detailing how they would manufacture it.

Teacher Self-Evaluation:

WEEK 3: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Area

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify the uses of area in real-life situations.
- Use digital devices to search for information on the uses of area in real-life situations.
- Acknowledge the uses of area in real-life situations.

Key Inquiry Question:
- Where is area applied in real-life situations?

Learning Resources:
- Top Scholar Maths
- Digital devices (tablets/laptops)
- Lesson notes

Organisation of Learning

Introduction (5 minutes)
1. Briefly review the previous lesson on the fundamental concepts of area (e.g., formulas for different shapes).
2. Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of the important applications of area in real life. Encourage students to share any personal experiences relating to area.

Lesson Development (30 minutes)

Step 1: Group Brainstorming (10 minutes)
- Divide the class into small groups.
- Ask each group to brainstorm and list as many real-life situations or professions where area is important.
- Provide prompts to help (e.g., agriculture, architecture, art, etc.).
- One representative from each group will share their findings.

Step 2: Online Research (10 minutes)
- Provide learners with digital devices.

- Instruct students to conduct online research to find more examples of how area is used in real life.
- Encourage them to take notes and find at least three sources or examples.
- Remind them to be critical of the information and identify reliable sources.

Step 3: Group Discussion (5 minutes)
- In their groups, have students discuss their findings and what they discovered during their online research.
- Encourage them to think critically about the importance of area in various contexts and refine their ideas.

Step 4: Class Presentation (5 minutes)
- Each group presents their findings to the class, highlighting interesting uses of area they discovered.
- Facilitate a short discussion after each presentation to allow for questions or additional comments from classmates.

Conclusion (5 minutes)
- Summarize key points discussed during the lesson and revisit the learning objectives to highlight what has been achieved.
- Conduct a brief interactive activity, such as a quiz or a "Think-Pair-Share" to reinforce the main topics.
- Preview the next session by suggesting they think about how area relates to volume as an upcoming topic or pose a question: "How does understanding area help in other mathematical concepts?"

Extended Activities:
1. Design Project: Ask students to create a floor plan for a dream home and calculate the area of each room they design.
2. Field Study: Have students measure the area of a local park or playground using the formulas they've learned, comparing real-life measurements to calculated values.
3. Art Integration: Challenge students to create a piece of art that utilizes area (e.g., tiling, mosaics) and calculate the total area used in their design.

Teacher Self-Evaluation:

WEEK 3: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Volume of Solids

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Identify the formula for calculating the volume of a triangular-based prism.
2. Work out the volume of triangular-based prisms.
3. Acknowledge the formula for calculating the volume of triangular-based prisms.

Key Inquiry Question:
- How do you determine the volume of a triangular-based prism?

Learning Resources:
- Top Scholar Maths pg 104-107
- Containers and objects (for practical activity)
- Teacher's Guide
- Models of triangular prisms

Organisation of Learning:

Introduction (5 minutes):
- Begin the lesson by reviewing the previous concepts covered, such as basic geometric shapes and their volumes.
- Guide learners to read pages 104-107 from Top Scholar Maths and discuss the key concepts that lead to understanding the volume of triangular-based prisms.

Lesson Development (30 minutes):

Step 1: Identifying Triangular Prisms (10 minutes)
- In pairs or small groups, learners will explore their surroundings (classroom, home) for items that resemble triangular prisms (e.g., tents, wedge-shaped containers).
- Each group shares one example with the class and briefly explains why it is a triangular prism.

Step 2: Discussing Prism Models (10 minutes)
- Using models of triangular prisms, have learners observe different types of triangular prisms and discuss their characteristics.
- Facilitate a guided discussion on the shape, base area, and height of the prism.

Step 3: Identifying the Volume Formula (5 minutes)
- Introduce the formula for the volume of a triangular prism:
$$ \text{Volume} = \text{Base Area} \times \text{Height} $$
- Explain how the base area is calculated:
$$ \text{Base Area} = \frac{1}{2} \times \text{base} \times \text{height of triangle} $$
- Ask learners to note down the formula.

Step 4: Calculating Volume (5 minutes)
- Provide examples of dimensions for triangular prisms and guide learners to apply the formula they have just learned to calculate the volume.
- Allow pairs to work on a few examples and share their calculations with the class for assessment and feedback.

Conclusion (5 minutes):
- Summarize the key points discussed in the lesson: the formula for volume, the characteristics of triangular prisms, and how to calculate their volume.
- Engage the class in a brief interactive activity, such as a quiz or a brief group discussion on a related question to reinforce the concepts.
- Briefly introduce the next topic to prepare learners for upcoming lessons, potentially about other types of prisms or volume-related concepts.

Extended Activities:
- Have learners create a 3D model of a triangular-based prism using craft materials and calculate the volume based on their model's dimensions.
- Encourage students to explore real-world applications of triangular prisms, such as in architecture, and present their findings in a group project.
- Assign practice problems from Top Scholar Maths or online geometry platforms to further their understanding and application of calculating the volume of triangular prisms.

Teacher Self-Evaluation:

WEEK 3: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Volume of Solids

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify the formula for calculating the volume of a rectangular-based prism.
- Work out the volume of rectangular-based prisms.
- Enjoy calculating the volume of rectangular-based prisms.

Key Inquiry Questions:
- How do you find the volume of a rectangular-based prism?

Learning Resources:
- Top Scholar Maths (pg 106-107)
- Teacher's Guide
- Chalkboard
- Model of a rectangular-based prism
- Various containers and objects resembling rectangular prisms

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson by asking students to share what they learned about measurements and units.
- Guide learners to read and discuss relevant content (pg 106-107) from the learning resources, emphasizing understanding key concepts about volume.

Lesson Development (30 minutes)

Step 1: Object Identification (10 minutes)
- In groups, learners will observe various containers and objects (provided by the teacher), identifying those that resemble rectangular prisms.
- Encourage discussion within groups about the characteristics of these objects.

Step 2: Formula Introduction (5 minutes)
- Guide the class in identifying the formula for calculating the volume of rectangular prisms:
\[

\text{Volume} = \text{Length} \times \text{Width} \times \text{Height}
\]
- Write the formula on the board and explain each component.

Step 3: Demonstration (5 minutes)
- Using the model of a rectangular-based prism, the teacher will illustrate how to measure the length, width, and height, followed by a demonstration on how to apply the formula in a real example.

Step 4: Hands-on Practice (10 minutes)
- In pairs or individually, learners will select a rectangular-based prism from the provided containers and measure its dimensions. They will apply the formula to calculate the volume and record their findings.
- Learners will share their results with their peers for assessment and feedback.

Conclusion (5 minutes)
- Summarize the key points and learning objectives achieved during the lesson.
- Conduct a brief interactive activity where students volunteers demonstrate a calculation on the board.
- Prepare learners for the next session by introducing upcoming topics, such as additional types of prisms or complex volume problems.

Extended Activities:
- Real-Life Application: Assign students to find everyday objects at home that are rectangular prisms and calculate their volumes. They can present their findings in the next class.
- Volume Challenges: Create a set of word problems involving the volume of rectangular prisms for homework to deepen their understanding.
- Prism Exploration: In groups, students can design a rectangular-based prism with given volume constraints, then share their designs and calculations with the class.

Teacher Self-Evaluation:

WEEK 4: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Volume of Solids

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Identify the formula for calculating the volume of square-based pyramids.
2. Calculate the volume of a square-based pyramid.
3. Enjoy working out the volume of square-based pyramids.

Key Inquiry Question:
- How do we calculate the volume of square-based pyramids?

Learning Resources:
- Top Scholar Maths (pg. 107-108)
- Video clips
- Digital devices (tablets or computers)
- Teacher's Guide
- Manillas (large sheets of paper)
- Nets of square-based pyramids
- Calculators

Organisation of Learning

Introduction (5 minutes)
- Begin with a short review of the previous lesson on volume.
- Ask students to volunteer any formulas they remember.
- Introduce the topic of square-based pyramids and highlight the importance of calculating their volume.

Lesson Development (30 minutes)

Step 1: Draw a Net of the Square-Based Pyramid (10 minutes)
- In groups, students will use manillas to draw the net of a square-based pyramid.
- Encourage creativity in designing the net and ensure they discuss the different parts of the pyramid.

Step 2: Explore Volume Calculation Videos (10 minutes)

- Provide students with digital devices where they can access specific video clips on calculating the volume of square-based pyramids.
- Instruct them to take notes on the key points and formulas presented in the videos.

Step 3: Identifying the Volume Formula (5 minutes)
- After watching the videos, guide each group to identify and write down the formula for the volume of a square-based pyramid:
$$ V = \frac{1}{3} \times base \, area \times height $$
- Discuss the significance of each component in the formula.

Step 4: Calculate Volumes (5 minutes)
- Assign each group different dimensions to use in calculating the volume, allowing them to apply the formula.
- Walk around to provide real-time feedback and support.

Conclusion (5 minutes)
- Summarize the key points discussed during the lesson, including the formula and calculation process for the volume of square-based pyramids.
- Conduct a brief interactive activity, such as a flash quiz or a quick group discussion, to reinforce the topics learned.
- Provide a preview of the next lesson, which will cover composite solids, and pose questions for them to think about: "How do different shapes interact when we talk about volume?"

Extended Activities:
- Volume Project: Students can create a model of a square-based pyramid using household materials and calculate its actual volume to compare with their theoretical calculations.
- Math Journal: Encourage students to keep a math journal where they can write down the formulas and include a section for reflections on what they learned during the lesson as well as questions they still have.

Teacher Self-Evaluation:

WEEK 4: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Volume of Solids

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Identify the formula for calculating the volume of a rectangular-based pyramid.
- Calculate the volume of rectangular-based pyramids.
- Acknowledge the formula and steps for calculating the volume of rectangular-based pyramids.

Key Inquiry Question:
- How do we calculate the volume of a rectangular-based pyramid?

Learning Resources:
- *Top Scholar Maths* (pages 109-110)
- Chalkboard & chalks
- Teacher's Guide

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on solid shapes and their properties.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the structure and properties of rectangular-based pyramids.

Lesson Development (30 minutes)

Step 1: Identifying the Formula (10 minutes)
- In small groups, students will brainstorm what they already know about pyramids and solids.
- The teacher will introduce the volume formula for a rectangular-based pyramid:
\[V = \frac{1}{3} \times \text{Base Area} \times \text{Height} \]
- Each group will write the formula on the board and discuss the meaning of each component (Base Area, Height).

Step 2: Demonstrating the Calculation (10 minutes)
- The teacher will illustrate how to apply the formula.

- Example: Calculate the volume of a pyramid with a base length of 4 cm, width of 3 cm, and height of 6 cm.
- Encourage students to ask questions.

Step 3: Practice Calculation in Pairs (5 minutes)
- Students will pair up and work with provided examples of rectangular-based pyramids.
- Each pair will calculate the volume of different pyramids based on given dimensions.

Step 4: Peer Sharing and Feedback (5 minutes)
- Groups will share their findings with the class.
- The teacher will facilitate a discussion to address any discrepancies and validate correct calculations.

Conclusion (5 minutes)
- Summarize the key points discussed: the formula for volume and the calculation steps.
- Conduct a brief interactive quiz or game to reinforce the main topics with questions that require the application of the formula.
- Prepare learners for the next session by previewing upcoming geometric shapes to study, such as cones and cylinders.

Extended Activities:
- Real-World Application Project: Have students research buildings or structures with pyramid shapes and calculate the volume based on specific dimensions, discussing their importance in design or architecture.
- Hands-On Activity: Create paper models of rectangular-based pyramids, and have students measure and calculate the volume of their models to connect theory with physical objects.

Teacher Self-Evaluation:

WEEK 4: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Volume of Solids

Specific Learning Outcomes:
- By the end of the lesson, learners should be able to:
- Identify the formula for calculating the volume of a triangular-based pyramid.
- Calculate the volume of a triangular-based pyramid.
- Enjoy working out the volume of triangular-based pyramids.

Key Inquiry Question(s):
- How do we calculate the volume of triangular-based pyramids?

Learning Resources:
- Top Scholar Maths, pg 109-110
- Teacher's Guide
- Chalkboard & chalk
- Model of a triangular-based pyramid

Organisation of Learning

Introduction (5 minutes)
- Begin with a quick review of the previous lesson on volume, asking students to share what they remember.
- Guide learners to read and discuss the relevant sections from the textbook, emphasizing the importance of understanding triangular-based pyramids.

Lesson Development (30 minutes)

Step 1: Introduction to the Formula (10 minutes)
- Present the formula for calculating the volume of a triangular-based pyramid:
$$\text{Volume} = \frac{1}{3} \times \text{Base Area} \times \text{Height}$$
- Explain each component of the formula: the base area (which can be calculated using the formula for the area of a triangle), and height.
- Use the model of a triangular-based pyramid to visualize these concepts as you explain.

Step 2: Group Discussion and Exploration (10 minutes)
- Divide students into small groups or pairs.

- Ask them to derive the base area formula for a triangle (e.g., $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$) and discuss its application in the volume formula.
- Have groups share their thoughts and findings with the class.

Step 3: Guided Calculation (5 minutes)
- Using the chalkboard, walk students through an example problem to calculate the volume of a triangular-based pyramid with specific dimensions (e.g., base of 4 cm, height of 3 cm).
- Show calculations step-by-step, inviting learners to contribute their ideas throughout the problem.

Step 4: Practice with Feedback (5 minutes)
- Provide learners with a quick worksheet or problems on the board to independently calculate the volume of a few triangular-based pyramids with different dimensions.
- Circulate around the room to assess their work, providing on-the-spot feedback and assistance as needed.

Conclusion (5 minutes)
- Summarize key points from the lesson, highlighting the importance and application of the volume formula for triangular-based pyramids.
- Conduct a quick interactive game (like a quiz) where students can answer questions about the formulas and volume calculations, reinforcing today's learning.
- Preview the next lesson's topic: "Exploring Other 3D Shapes and Their Volumes" and pose a few questions for them to consider at home.

Extended Activities:
- Challenge students to create a model of a real-life object that resembles a triangular-based pyramid and calculate its volume.
- Encourage learners to research additional volume formulas for other 3D shapes and prepare short presentations for the next class.
- Assign an online interactive geometry tool or app where students can manipulate 3D shapes and visualize changes in volume based on different dimensions.

Teacher Self-Evaluation:

WEEK 4: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub-Strand: Volume of Solids

Specific Learning Outcomes:
- Identify the formula for calculating the volume of a cone.
- Work out the volume of a cone in real-life situations.
- Acknowledge the formula for calculating the volume of a cone.

Key Inquiry Question(s):
- How do we find out the volume of cones?

Learning Resources:
- Top Scholar Maths (pg 110-112)
- Chalkboard & charts
- Manilla papers & pencils
- Digital devices
- Video clips

Organisation of Learning:

Introduction (5 minutes)
- Begin the class with a brief review of the previous lesson focused on three-dimensional shapes.
- Ask students to share what they remember about cylinders and other solids, leading into a discussion about cones.
- Engage learners in reading and discussing the relevant content from pages 110-112 of the learning resource, emphasizing formula identification.

Lesson Development (30 minutes)

Step 1: Draw a Net or Model (10 minutes)
- Divide students into pairs or small groups.
- Distribute manila papers and ask each group to draw a net of a cone and cut it out. Discuss the components of the cone: radius, height, and base.

Step 2: Explore Video Resource (10 minutes)
- Guide students to use digital devices to watch a video about calculating the volume of a cone. Ensure the video clearly explains the formula and demonstrates examples.

- After watching, discuss the video as a class, allowing students to ask questions for clarity.

Step 3: Identify the Formula (5 minutes)
- Once the video has been reviewed, write the volume formula for a cone on the board:
$$ V = \frac{1}{3} \pi r^2 h $$
- Ask students to break down each part of the formula and what it represents (radius, height, and π value).

Step 4: Apply the Formula (5 minutes)
- Provide students with different scenarios or real-life objects resembling cones (like ice cream cones, traffic cones, etc.) and ask them to calculate the volume using the formula.
- Encourage groups to share their answers and reasoning with the class.

Conclusion (5 minutes)
- Summarize the key points covered in the lesson, reiterating the formula for the volume of a cone and its application.
- Conduct a quick interactive quiz where students can answer the volume of cones based on different dimensions you provide. Use thumbs up/down or online polling for engagement.
- Provide a brief teaser about the next lesson (e.g., exploring the volume of other solids like cylinders and spheres) and pose a question for them to think about.

Extended Activities:
- Encourage students to bring in a cone-shaped object from home and calculate its volume using measurements they take.
- Assign a project where students research real-life applications of cones, such as in architecture or manufacturing, and present their findings to the class.
- Provide online interactive simulations where learners can adjust the height and radius of cones to see how the volume changes.

Teacher Self-Evaluation:

WEEK 4: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Volume of Solids

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify the formula and steps for calculating the volume of frustums.
- Determine the volume of a frustum in real-life situations.
- Show interest in learning how to determine the volume of a frustum.

Key Inquiry Question:
- How do we determine the volume of a frustum?

Learning Resources:
- Textbook: Top Scholar Maths pg 112-115
- Video clip on calculating frustum volume
- Digital devices (tablets/computers)
- Chalkboard & chalk
- Models of frustums (e.g., plastic or cardboard)
- Calculators

Organisation of Learning:

Introduction (5 minutes)
- Begin with a brief review of the previous lesson on volume.
- Engage learners in a discussion about the properties of 3D shapes, especially focusing on cones and cylinders.
- Introduce the concept of a frustum and provide examples from real life (e.g., a truncated cone shape like a lampshade) to spark interest.

Lesson Development (30 minutes)

Step 1: Model Creation (10 minutes)
- In small groups, learners will create a physical model of a frustum using locally available materials such as paper cups or cardboard pieces.
- Encourage learners to visualize how a frustum relates to cones and cylinders by stacking and cutting shapes.

Step 2: Video Exploration (5 minutes)
- Guide learners to use digital devices to access a video clip that explains how to calculate the volume of a frustum.
- After viewing, have groups discuss what they learned and how the video relates to their models.

Step 3: Formula Identification (10 minutes)
- Lead a discussion on the formula for the volume of a frustum:
- Have learners identify each component of the formula using their models.

Step 4: Volume Calculation Practice (5 minutes)
- Provide learners with different frustum problems (real-world examples) to work on collaboratively.
- Each group will calculate the volume using the formula discussed.
- Encourage learners to share their answers with the class for feedback.

Conclusion (5 minutes)
- Summarize key points from the lesson: the definition of a frustum, the volume formula, and its application in real life.
- Conduct a brief interactive quiz or game to reinforce key concepts.
- Preview the next lesson on surface area of solids and pose questions for students to consider, such as: “How does knowing the volume help us find the surface area?”

Extended Activities:
- Frustum Art Project: Learners can create artistic designs using frustums and calculate their volumes.
- Real-Life Investigation: Learners can find objects around their homes that resemble frustums and measure them to determine their volumes.
- Research Task: Investigate and present on how frustums are used in engineering or architecture, linking math to real-world applications.

Teacher Self-Evaluation:

WEEK 5: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Volume of solids

Specific Learning Outcomes:
- By the end of the lesson, learners should be able to:
- Identify the formula for calculating the volume of a sphere.
- Calculate the volume of a sphere in real-life situations.
- Acknowledge the formula for working out the volume of spheres.

Key Inquiry Question(s):
- How do we calculate the volume of spheres?

Learning Resources:
- Top Scholar Maths pg 115-116
- Video clip on calculating the volume of a sphere
- Digital devices (tablets/laptops)
- Chalkboard & chalks
- Calculators

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on three-dimensional shapes.
- Guide learners to read and discuss relevant content from the learning resources, focusing on the volume of a sphere and connecting it to their prior knowledge.

Lesson Development (30 minutes)

Step 1: Understanding the Formula (10 minutes)
- Use the chalkboard to write the formula for the volume of a sphere:
\[V = \frac{4}{3} \pi r^3 \]
- Explain each component of the formula (V = volume, r = radius, π (pi) is approximately 3.14).
- Engage learners in a discussion about the significance of the radius and how it is used in this formula.

Step 2: Video Exploration (5 minutes)
- Split students into pairs or small groups.

- Assign each group to use digital devices to search for and watch a video on calculating the volume of a sphere.
- Encourage groups to take notes on key points from the video.

Step 3: Hands-On Practice (10 minutes)
- Provide learners with real-life scenarios (e.g., finding the volume of a basketball, a globe, or a water balloon).
- Have each group calculate the volume of these objects using the formula.
- Provide different radii for each scenario to enhance practice.

Step 4: Sharing Findings (5 minutes)
- Ask each group to present their findings to the class.
- Encourage peer assessment by having other groups provide feedback on the presented calculations and clarify any misunderstandings.

Conclusion (5 minutes)
- Summarize the key points discussed in the lesson:
- The formula for volume of a sphere and its components.
- Real-life applications of the formula.
- Conduct a brief interactive quiz or activity to reinforce the main topics: for example, "What would happen to the volume if the radius doubled?"
- Preview the next session by hinting at calculating the volume of other solids (like cylinders or cones) and encourage students to consider how these concepts are interconnected.

Extended Activities:
- Volume Project: Have students measure the radius of household spherical objects (like balls or fruit) and calculate their volumes. They can create a poster displaying their findings, including images and calculations.
- Research Assignment: Assign learners to research the use of spheres in real-world applications, such as in engineering, sports, and nature, and present their findings to the class.

Teacher Self-Evaluation:

WEEK 5: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Volume of solids

Specific Learning Outcomes:
- Identify the use of volume of solids in real-life situations.
- Use digital devices to find out the use of volume of solids in real-life situations.
- Promote the use of volume and capacity of different containers in real-life situations.

Key Inquiry Question(s):
- How do we use the volume of solids in real-life situations?

Learning Resources:
- Top Scholar Maths
- Digital devices (tablets, computers, or smartphones)

Organisation of Learning:

Introduction (5 minutes)
- Review: Begin the lesson by quickly revisiting the previous topic on measurements, focusing on the importance of measuring length and capacity.
- Discussion: Prompt students to think about where they might encounter volume in their daily lives (e.g., filling a water bottle, packing boxes). Encourage them to share their thoughts as they relate to volume.

Lesson Development (30 minutes)

Step 1: Group Brainstorming (10 minutes)
- Activity: Divide students into small groups (4-5 per group). Each group will brainstorm and list at least five real-life applications of the volume of solids (e.g., construction, food storage, shipping).
- Outcome: Groups will create a mind map on their digital devices to illustrate their ideas.

Step 2: Digital Research (10 minutes)
- Activity: Students will use digital devices to research a specific real-life example where volume is vital, such as the design of a swimming pool or packaging for a product. Encourage them to note how these examples use volume in calculations.
- Outcome: Each group will write a brief summary of their findings.

Step 3: Group Discussion (5 minutes)
- Activity: Groups will convene to discuss their research findings, focusing on how volume affects decisions in practical scenarios.
- Outcome: Each group prepares to present their findings to the class.

Step 4: Class Presentation (5 minutes)
- Activity: Each group will present their mind map and research summary to the class. Encourage students to ask questions and engage in discussions after each presentation.
- Outcome: Students improve presentation skills and learn about various applications of volume from their peers.

Conclusion (5 minutes)
- Summary: Recap the key points about the real-life applications of volume discussed during presentations.
- Interactive Activity: Conduct a fast-paced quiz using an online platform (like Kahoot) to reinforce what students learned about volume applications.
- Preparation: Preview the next lesson, which will cover calculating the volume of different solids (prism and cylinder), and pose a question: “How can we calculate the volume of a container before we fill it?”

Extended Activities:
- Home Project: Have students find and measure three different containers at home (e.g., a box, a bottle, and a can). Ask them to calculate their volumes and think about how these are used in daily life.
- Research Task: Students could create a digital poster or presentation about a real-world architectural project that considers volume in its design (e.g., sports stadium, aquarium).
- Math Application: Encourage students to bring in products packaged in unique containers and calculate their volumes to discuss efficiency in packaging.

Teacher Self-Evaluation:

WEEK 5: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub-Strand: Mass, Volume, Weight, and Density

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Identify the instruments and tools used in weighing materials in the locality.
- Convert units of mass from one form to another in different situations.
- Enjoy converting units of mass from one form to another.

Key Inquiry Question(s):
- How do you weigh materials and objects?
- How do you convert the different units of mass from one form to another?

Learning Resources:
- Top Scholar Maths (pages 117-119)
- Conversion charts
- Electronic balance
- Different materials and objects for weighing
- Lesson notes
- Teacher's Guide

Organisation of Learning:

Introduction (5 minutes)
1. Begin with a review of the previous lesson on measurements and their applications in real life.
2. Facilitate a discussion where students share their experiences with weighing different objects and introduce key concepts related to mass and its measurement.

Lesson Development (30 minutes)

Step 1: Identifying Instruments (10 minutes)
- Activity: In small groups, learners identify various instruments and tools they may have seen or used for weighing (e.g., kitchen scale, balance scale, electronic balance).
- Discussion: Relate these tools to consumer awareness and protection. Why is it important to use accurate instruments?

Step 2: Practical Weighing (10 minutes)

- Activity: Distribute several objects/materials for weighing and assign each group an electronic balance.
- Task: Each group weighs their assigned materials and records the weights. Encourage accuracy in reading the instruments.

Step 3: Conversion Exploration (5 minutes)
- Activity: Students use conversion charts to familiarize themselves with different units of mass (grams, kilograms, pounds, etc.).
- Discussion: Groups share how to convert the weights they collected into another unit (e.g., grams to kilograms), and discuss real-life applications of these conversions.

Step 4: Unit Conversion Practice (5 minutes)
- Activity: Provide students with several practice problems that require converting units of mass in different contexts (e.g., converting the mass of ingredients in a recipe).
- Assessment: Circulate and provide feedback as groups work through the problems, clarifying misconceptions as needed.

Conclusion (5 minutes)
- Summarize the key concepts learned about weighing, tools, and unit conversions.
- Conduct a brief interactive activity like a quiz or a quick-fire question round to reinforce learning.
- Provide a preview of the next lesson on volume and how it connects to the topics covered today.

Extended Activities:
- Challenge students to research and present on different historical methods of weighing materials.
- Assign a project where they create a recipe that requires unit conversions of mass, encouraging practical application in cooking or baking.
- Introduce an online game or app that focuses on practicing measurements and conversions.

Teacher Self-Evaluation:

WEEK 5: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Mass, Volume, Weight, and Density

Specific Learning Outcomes:
By the end of the lesson, learners will be able to:
- Define the terms mass and weight and give their SI units.
- Relate mass and weight in real-life situations.
- Acknowledge the relationship between mass and weight.

Key Inquiry Question:
- What is the relationship between mass and weight?

Learning Resources:
- Top Scholar Maths (pg 119-120)
- Weighing tools and instruments
- Different objects and materials
- Teacher's Guide
- Lesson notes

Organisation of Learning

Introduction (5 minutes)
- Review the previous lesson (e.g., dimensions of measurements, basic definitions).
- Guide learners to read and discuss relevant content from the resources, emphasizing defining mass and weight.

Lesson Development (30 minutes)

Step 1: Definitions and SI Units (10 minutes)
- In groups, learners will use digital devices to search for the definitions of mass and weight and their SI units (Mass in kilograms, Weight in Newtons).
- Each group will note down their definitions in their notebooks and prepare to share with the class.

Step 2: Discussion on Relationship (5 minutes)
- Once definitions are shared, lead a class discussion on the relationship between mass and weight.

- Address questions like, "How does the weight of an object change based on its location, like on the Moon versus Earth?"

Step 3: Practical Measurement (10 minutes)
- Using selected instruments (scales, spring scales), groups will measure the masses of various objects (e.g., fruits, books). Each group will record the mass in kilograms.
- Demonstrate how to calculate weight using the formula:

Step 4: Finding Weight and Vice Versa (5 minutes)
- Encourage students to swap roles within groups, so some measure while others calculate the weight.
- Challenge learners to find the mass of a given weight using rearranging the formula:

Conclusion (5 minutes)
- Summarize key concepts: definitions of mass and weight, SI units, and the relationship illustrated through practical activity.
- Conduct a brief interactive quiz to reinforce learning (e.g., True/False questions about mass and weight).
- Prepare learners for the next session, hinting at upcoming topics such as density and volume.

Extended Activities
- Research Project: Assign learners to choose an object and research its mass, weight, and how they vary in different locations (Earth vs. Moon).
- Measurement Hunt: Have learners conduct a “measurement hunt” at home where they gather different items, measure them, and record their mass and weight (with parental guidance if needed).
- Density Exploration: Introduce the concept of density and have students calculate the density of various objects after measuring their mass and volume in a follow-up lesson.

Teacher Self-Evaluation:

WEEK 5: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Mass, Volume, Weight, and Density

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Explain how to determine mass, volume, and density in different situations.
- Determine mass, volume, and density in different situations.
- Enjoy carrying out activities relating to mass, volume, and density using different substances.

Key Inquiry Question:
- How do you determine mass, volume, and density in different situations?

Learning Resources:
- Top Scholar Maths (pg 120-121), calculators, various objects (coins, stones, padlock), measuring cylinders, electronic balance, water, and a recording table.

Organisation of Learning:

Introduction (5 minutes):
- Review the previous lesson briefly, asking students to share key takeaways.
- Provide a short overview of mass, volume, and density, and guide learners to read relevant sections from the learning resources.
- Emphasize real-world applications of these concepts.

Lesson Development (30 minutes):
Step 1: Discussion (10 minutes)
- In small groups, have students discuss how mass, volume, and density can be determined.
- Ask guiding questions: "What tools do we need?" and "What real-life situations require these calculations?"

Step 2: Practical Measurements (10 minutes)
- Provide each group with measuring cylinders, coins, stones, and an electronic balance.
- Instruct students to measure the mass of different objects and use displacement in water to find their volume if necessary.
- Record their findings in a table.

Step 3: Calculating Density (5 minutes)

- Introduce the formula for density: Density = Mass / Volume.
- Have each group calculate the density of the objects they measured using their recorded data.

Step 4: Sharing Findings (5 minutes)
- Invite groups to share their findings with the class.
- Discuss any discrepancies in data and encourage students to reflect on what they learned from the exercise.

Conclusion (5 minutes):
- Summarize the key points covered, such as the definitions and calculations of mass, volume, and density.
- Conduct a short interactive quiz where students can answer questions based on what they learned today.
- Provide a brief overview of the next session, hinting at how these concepts apply to other scenarios like buoyancy or material properties.

Extended Activities:
- Encourage students to conduct a home experiment where they measure the density of various household liquids (like oil, water, and vinegar) and record their observations.
- Assign a project where students select a common object and research how understanding its density can influence its use or manufacturing process (e.g., why certain materials are lighter or heavier).

Teacher Self-Evaluation:

WEEK 6: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Mass, Volume, Weight and Density

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Identify the formula for determining density of objects given mass and volume.
- Calculate the density of different objects given the mass and volume in various situations.
- Enjoy finding the density of objects given mass and volume.

Key Inquiry Question(s):
- How do you determine the density of objects given the mass and volume?

Learning Resources:
- Top Scholar Maths pg 121-122
- Calculators
- Teacher's Guide
- Chalkboard and chalks
- Illustration chart

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson about basic measurements (mass and volume).
- Guide learners to read and discuss the relevant content from the learning resources, emphasizing a clear understanding of density.

Lesson Development (30 minutes)

Step 1: *(Discuss the Concept of Density)*
- Introduce the concept of density and its importance in real-life situations.
- Ask students to think of objects with different densities (e.g., oil and water).

Step 2: *(Identify the Density Formula)*
- Present the formula for density: Density = Mass/Volume.
- Write it on the chalkboard and talk through the components - mass (grams) and volume (liters or cubic centimeters).

Step 3: *(Practice Calculation)*
- Provide students with several examples of objects (with given mass and volume).
- In groups, have them calculate the density of these objects using the formula.
- Encourage collaboration and discussion among group members.

Step 4: *(Share and Assess Findings)*
- Each group shares their findings with the class.
- Provide feedback and assess their understanding while discussing any discrepancies in calculations.

Conclusion (5 minutes)
- Summarize the key points covered: the density formula and how to calculate it.
- Conduct a quick interactive activity: ask students to raise their hands if they can remember the formula or give an example of an object with a specific density.
- Briefly preview the next lesson, which will cover practical applications of density (e.g., buoyancy).

Extended Activities:
- Density Lab: Have students collect different household liquids and solids at home (e.g., sugar water, oil, salt) and measure their mass and volume to determine density.
- Research Project: Assign a short research project on the role of density in natural phenomena (e.g., why some objects float while others sink).

Teacher Self-Evaluation:

WEEK 6: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Mass, Volume, Weight, and Density

Specific Learning Outcomes:
- By the end of the lesson, the learner should be able to:
- Identify the formula for determining the mass of objects given volume and density.
- Calculate the mass of objects given volume and density.
- Enjoy calculating the mass of objects given volume and density.

Key Inquiry Question(s):
- How do we find mass given the volume and density?

Learning Resources:
- Top Scholar Maths pg 122
- Calculators
- Teacher's Guide
- Chalks and Chalkboard

Organisation of Learning:

Introduction (5 minutes):
- Review the previous lesson by asking students to recall what they learned about mass, volume, and density.
- Guide learners to read and discuss relevant content from page 122 of the Top Scholar Maths book, emphasizing the understanding of density and its relation to mass and volume.

Lesson Development (30 minutes):

Step 1: Identify the Formula (10 minutes)
- Explain the relationship between mass, volume, and density using the formula:
\(\text{Mass} = \text{Density} \times \text{Volume} \)
- Annotate the formula on the chalkboard and ensure all students understand each component.

Step 2: Discuss Examples (10 minutes)
- In pairs, provide students with different volumes and densities. Ask them to discuss and calculate the mass together using the formula.

- Allow students to share one or two example problems with the class, ensuring they articulate their thought processes.

Step 3: Calculate Mass (5 minutes)
- Distribute a set of problems for students to solve individually or in their pairs using calculators. Encourage them to apply the formula they learned.
- Circulate the room to assist and provide feedback on their calculations.

Step 4: Share Findings (5 minutes)
- Have a few students share their calculations and explain how they arrived at their answers.
- Facilitate a brief discussion on different scenarios where mass would vary based on density and volume.

Conclusion (5 minutes):
- Summarize the key points learned about mass, density, and volume.
- Conduct an interactive activity, such as a quick quiz where students answer questions or solve problems on mini whiteboards.
- Prepare learners for the next session by previewing the topic of weight and how it differs from mass.

Extended Activities:
1. Density Experiment: Encourage students to find the mass and volume of various objects at home and calculate their densities. They can present their findings in the next class.
2. Graphing Task: Have students collect data on the mass and volume of classroom objects and graph the result, showcasing the relationship between mass and volume visually.
3. Research Assignment: Students can prepare a brief presentation on how different industries apply the concept of mass, volume, and density, such as in shipping or cooking.

Teacher Self-Evaluation:

WEEK 6: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Mass, Volume, Weight, and Density

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Identify the formula for determining volume given the mass and density.
2. Calculate the volume given mass and density.
3. Enjoy finding the volume given mass and density.

Key Inquiry Question:
- How do you determine the volume of a given mass and density?

Learning Resources:
- Top Scholar Mathematics pg 123
- Teacher's Guide
- Calculators
- Chalkboard and Chalks

Organisation of Learning

Introduction (5 minutes)
- Review the previous lesson on density and its relationship to mass and volume.
- Guide learners to read and discuss relevant content from the learning resources, focusing on the importance of understanding the formula for volume**.**

Lesson Development (30 minutes)

Step 1: Understanding the Formula (10 minutes)
- Introduce the formula for volume:
\[V = \frac{m}{d} \]
where \(V \) is volume, \(m \) is mass, and \(d \) is density.
- Discuss the components of the formula and their units (e.g. grams for mass, grams per cubic centimeter for density, resulting volume in cubic centimeters).

Step 2: Guided Practice (10 minutes)
- Demonstrate a sample problem on the chalkboard:
- "Find the volume of a substance that has a mass of 150 grams and a density of 3 g/cm^3."

- Work through the calculation together:
$$ V = \frac{150 \, \text{g}}{3 \, \text{g/cm}^3} = 50 \, \text{cm}^3 $$
- Engage learners by asking guiding questions throughout the problem-solving process.

Step 3: Group Work (5 minutes)
- Divide learners into pairs or small groups.
- Provide each group with different mass and density values and have them calculate the volume using the formula.
- Example problems:
1. Mass = 240g, Density = 4g/cm³
2. Mass = 180g, Density = 2g/cm³

Step 4: Sharing Findings (5 minutes)
- Allow each group to present at least one problem and solution to the class.
- Provide feedback on their calculations and clarify any misunderstandings.

Conclusion (5 minutes)
- Summarize the key points covered in the lesson: the formula $V = \frac{m}{d}$, how to apply it, and the importance of understanding volume in real-world contexts.
- Conduct a brief interactive activity, such as a quick quiz or a question-and-answer session, to reinforce the main topics.
- Prepare learners for the next session by hinting at how volume relates to other measurements and asking what other contexts they think density might be important in.

Extended Activities
- Investigate different substances around the home (e.g., liquids or powders) and calculate their volume based on mass and density. Submit a report summarizing findings.
- Create a density and volume chart for various common materials (water, oil, etc.) and compare their properties scientifically.

Teacher Self-Evaluation:

WEEK 6: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Mass, Volume, Weight, and Density

Specific Learning Outcomes:
- Identify the uses of density in daily life.
- Apply density to real-life situations.
- Recognize the use of density in daily life.

Key Inquiry Question(s):
- How can we apply density to real-life situations?

Learning Resources:
- Top Scholar Mathematics
- Digital devices
- Lesson notes and charts

Organisation of Learning:

Introduction (5 minutes)
- Review the key concepts from the previous lesson about mass, volume, and weight.
- Guide learners to read selected sections from the learning resources that discuss density, highlighting its importance in daily life.

Lesson Development (30 minutes)

Step 1: Brainstorming (5 minutes)
- In small groups, students brainstorm and list examples of where they think density is important in everyday life (e.g., floating and sinking, cooking, construction, etc.).
- Each group will share their ideas with the class, fostering discussion.

Step 2: Research (10 minutes)
- Using digital devices, students will search for specific applications or incidents where density plays a crucial role in real-life situations (e.g., how density affects the buoyancy of a boat, how density impacts weather patterns, etc.).
- Each group takes notes on their findings.

Step 3: Group Discussion (10 minutes)

- Students return to their groups to review their research findings and discuss the applications link to the concept of density.
- They will prepare a brief presentation outlining their findings to share with the class.

Step 4: Presentations (5 minutes)
- Each group presents their findings to the class, explaining how density is applied in their chosen examples.
- Encourage questions and interaction from classmates after each presentation.

Conclusion (5 minutes)
- Summarize key points covered in the lesson, ensuring to reiterate how density is relevant in various real-life scenarios.
- Conduct a quick interactive wrap-up activity (like a quiz or a short game) to reinforce the main topics discussed.
- Preview the next session's content, prompting students to think about other properties and measurements they encounter.

Extended Activities:
- Density Experiment: Ask students to conduct a simple experiment at home or in class where they can observe density by layering different liquids (e.g., honey, water, and oil) and discussing why they separate in layers.
- Density in the News: Have students find an article or news report that discusses an application of density in some field (e.g., environmental science, engineering, etc.) and present it in the next class.
- Creative Project: Allow students to create a visual poster or infographic showcasing the applications of density in daily life, which can be displayed in the classroom.

Teacher Self-Evaluation:

WEEK 6: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Time, Distance, and Speed

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify activities that involve measuring distances and time.
- Conduct activities that involve measuring distances and time.
- Engage in fun activities involving measuring distances and time.

Key Inquiry Questions:
- How do we observe speed in daily activities?
- Which day-to-day activities involve measuring distances and time?

Learning Resources:
- Top Scholar Mathematics pg 124
- Stopwatches
- School Field
- Recording Tables
- Calculators

Organisation of Learning:

Introduction (5 minutes)
1. Review of Previous Lesson: Briefly recap the last lesson focusing on measurements and introduce today's topic on time, distance, and speed.
2. Discussion: Engage learners in reading from the mathematics textbook and encourage them to discuss how distances and time are measured in everyday life.

Lesson Development (30 minutes)

Step 1: Brainstorming Activities (10 minutes)
- Group Work: Divide students into small groups. Instruct them to brainstorm everyday activities where they measure distances and time (e.g., running, biking, driving).
- Presentation: Each group shares their ideas with the class.

Step 2: Practical Measurement Activity (10 minutes)

- Field Activity: Take students to the school field. Set up a running track event. Explain how to measure time and distance effectively while allocating roles: some students will run, while others will time and measure distances.
- Recording Data: Let students record the distances covered and the time taken for each participant in a recording table.

Step 3: Calculate Speed (5 minutes)
- Using the Formula: Introduce the formula for speed (Speed = Distance ÷ Time). Guide students in their groups to calculate the speed of each participant using the data they collected.
- Discussion of Results: Have each group share their calculated speeds and discuss any discrepancies they noticed.

Step 4: Reflect on Learning (5 minutes)
- Class Discussion: Reflect on the activities performed. Ask students how measuring distances and time can apply to their daily lives. Encourage them to think about the importance of these measurements in various contexts (e.g., sports, travel).

Conclusion (5 minutes)
- Summarize Key Points: Recap the main concepts learned: identifying activities, measurement techniques, and calculating speed.
- Interactive Activity: Conduct a quick quiz or game to reinforce the main topics discussed during the lesson.
- Preview Next Session: Briefly introduce what will be covered in the next class, such as deeper applications of speed in different fields (e.g., physics, sports).

Extended Activities:
- Research Project: Assign a project where learners research different methods of measuring distances and time in various professions (e.g., athletics, transportation).
- Creative Task: Have students measure the speed of a rolling object (like a toy car) down an incline and compare it to their running speeds, extending the learning on practical applications of speed measurements.

Teacher Self-Evaluation:

WEEK 7: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Time, Distance, and Speed

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Identify the formula for calculating speed in km/h in real life situations.
2. Work out speed in km/h in real life situations.
3. Appreciate the use of time and distance in real life situations.

Key Inquiry Question:
- How do we find speed in km/h**?**

Learning Resources:
- Top Scholar Maths pages 124-125
- Calculators
- Teacher's Guide
- Charts

Organisation of Learning:

Introduction (5 minutes):
- Begin with a brief review of the previous lesson, focusing on distance and time.
- Guide learners to read selected content from the learning resources on speed.
- Discuss the importance of understanding speed in everyday contexts (e.g., driving, traveling by train).

Lesson Development (30 minutes):

Step 1: Discussing Distance and Time (10 minutes)
- In groups, have students discuss the relationship between distance, time, and how they affect speed.
- Ask each group to brainstorm real-life scenarios where they have needed to measure distance and time.

Step 2: Identifying the Speed Formula (5 minutes)
- Guide students to uncover and write the formula for speed (Speed = Distance ÷ Time) and ensure they understand each component.

- Provide examples of how to express distance in kilometers and time in hours.

Step 3: Applying the Speed Formula (10 minutes)
- Use real-life examples (e.g., calculate the speed of a car that travels 150 km in 2 hours).
- In groups, learners will solve several problems using the formula for speed. Encourage peer teaching if some students grasp the formula faster.

Step 4: Sharing Findings and Feedback (5 minutes)
- Have each group present their calculations and the scenarios they used.
- Provide formative feedback and facilitate a class discussion on various applications of speed.

Conclusion (5 minutes):
- Summarize the key points covered during the lesson, including the formula for speed and how to apply it to real-life situations.
- Conduct a brief interactive quiz or game (e.g., Kahoot or a quick round of verbal questions) to reinforce the main topics and engage all learners.
- Preview the next lesson where they will learn about acceleration and how it relates to speed.

Extended Activities:
- Project: Ask students to track their own travel (walking or biking) for a week. Have them measure distance and time taken for each trip and calculate their average speeds.
- Research Assignment: Students can explore different modes of transport (car, bike, train) and compare their speeds. Why do they differ?
- Speed Record Game: A fun outdoor activity where students measure the time it takes to run a certain distance. They can compare speeds with each other.

Teacher Self-Evaluation:

WEEK 7: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Time, Distance, and Speed

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Identify the formula for calculating speed in meters per second (m/s).
2. Apply the formula to calculate speed in real-life situations.
3. Enjoy working through speed calculations in practical examples.

Key Inquiry Question:
- How do you find speed in m/s?

Learning Resources:
- Top Scholar Maths pg 125
- Teacher's Guide
- Chalkboard and chalk
- Chart (showing the formula for speed)
- Calculators

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson focused on distance and time.
- Introduce the concept of speed and engage students in discussing how speed affects daily activities (e.g., driving, running).
- Ask students if they can think of situations where speed is important.

Lesson Development (30 minutes)

Step 1: Introduce the Speed Formula (10 minutes)
- Present the formula for speed:
- Write this formula on the chalkboard and explain each component.
- Discuss units: Distance in meters (m), Time in seconds (s), hence Speed in m/s.

Step 2: Example Calculation (5 minutes)
- Provide a real-life scenario: "A car travels 150 meters in 5 seconds. What is its speed?"
- Guide students through the calculation together:

- Emphasize the importance of using the correct units.

Step 3: Group Work on Real-Life Situations (10 minutes)
- Divide the class into small groups and give each group a different real-life scenario involving speed (e.g., a runner, a cyclist, a train).
- Each group will calculate speed using the provided distance and time, using calculators if needed.

Step 4: Sharing Findings and Feedback (5 minutes)
- Have each group present their scenario and calculated speed to the class.
- Encourage peer feedback and address any misunderstandings or errors in calculations.

Conclusion (5 minutes)
- Summarize key points: the formula for speed, how to apply it, and its relevance to real-life situations.
- Conduct a quick interactive quiz where students answer questions about the formula and real-life speed examples.
- Preview the next lesson: "Next week, we will explore how speed can affect time and distance relationships."

Extended Activities:
1. Speed Project: Ask students to track their walking or cycling speed over a week, measuring different distances and times, and presenting their findings in class.
2. Speed Comparisons: Research different modes of transport (cars, trains, planes) and compare their speeds and travel times for a specific distance.
3. Speed Games: Create distance versus time graphs using their calculated speeds and analyze trends in their speed data.

Teacher Self-Evaluation:

WEEK 7: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Time, Distance, and Speed

Specific Learning Outcomes:
- By the end of the lesson, learners should be able to:
- Identify the formula for calculating average speed in real-life situations.
- Work out average speed in real-life situations.
- Acknowledge the formula for finding average speed.

Key Inquiry Question(s):
- How do we calculate the average speed?

Learning Resources:
- Top Scholar Maths (pg 126-128)
- Teacher's Guide
- Chalkboard and chalks
- Calculators

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson briefly.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of key concepts of time, distance, and speed.

Lesson Development (30 minutes)

Step 1: Discuss Real-Life Examples (10 minutes)
- In groups, learners discuss various situations where they encounter speed. Examples could include traveling by car, running, or cycling.
- Encourage learners to share their experiences and consider factors that may affect speed (e.g., traffic, fatigue).

Step 2: Introduce the Formula (10 minutes)
- Introduce the formula for average speed:
- Illustrate the formula on the chalkboard and guide learners to note it in their notebooks.
- Provide examples to demonstrate how to apply the formula using realistic scenarios.

Step 3: Guided Practice (5 minutes)
- Present a word problem on the chalkboard that involves calculating average speed.
- Guide the learners step-by-step to solve the problem with the class using the formula introduced.

Step 4: Independent Practice (5 minutes)
- Distribute practice problems and ask learners to calculate average speed based on real-life scenarios provided (e.g., traveling 120 km in 2 hours).
- Circulate the room to provide support as needed.

Conclusion (5 minutes)
- Summarize key points made during the lesson: the formula for average speed and its application.
- Conduct a brief interactive quiz where learners can answer how they would calculate average speed in different scenarios.
- Preview the next session focusing on acceleration and how it relates to speed.

Extended Activities:
- Encourage learners to keep a travel diary for one week, measuring the distance and time taken for various activities (walking, biking, etc.), and then calculate their average speed for each activity.
- Create a project where learners analyze different modes of transportation and compare their average speeds, considering factors like time efficiency in different contexts (e.g., car vs. bike).

Teacher Self-Evaluation:

WEEK 7: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub-Strand: Time, Distance, and Speed

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Identify the formula for determining velocity in real-life situations.
2. Determine velocity in real-life situations.
3. Recognize the difference between speed and velocity.

Key Inquiry Questions:
- What is the difference between velocity and speed?
- How do we determine velocity?

Learning Resources:
- Top Scholar Maths, pg. 128-129
- Calculators
- Teacher's Guide
- Chalkboard and chalks

Organisation of Learning:

Introduction (5 minutes)
1. Review the previous lesson on speed and introduce today's focus on velocity.
2. Guide learners to read relevant content from pages 128-129 of the textbook. Discuss key concepts and terms, emphasizing the difference between speed and velocity.

Lesson Development (30 minutes)

Step 1: Understanding Speed vs. Velocity (10 minutes)
- Gather students in small groups to discuss the characteristics of speed (scalar) and velocity (vector).
- Use real-life examples (e.g., a car moving at 60 km/h vs. a car moving 60 km/h north) to illustrate the concept.
- Have each group share their definitions and examples with the class.

Step 2: Introducing the Formula for Velocity (10 minutes)
- Present the formula for velocity:

- Work through a basic example on the chalkboard (e.g., If a cyclist travels 30 km in 2 hours, what is their velocity?).
- Ask students to compute velocity using different examples from their group discussion.

Step 3: Real-life Situations (5 minutes)
- Assign real-life scenarios where students need to determine velocity (e.g., a runner covering 400 meters in 50 seconds).
- Let students calculate and discuss their findings in pairs or small groups.

Step 4: Sharing Findings and Class Discussion (5 minutes)
- Invite groups to present their calculations and discuss any differences in their findings.
- Highlight any common misconceptions about speed and velocity, and clarify these with the class.

Conclusion (5 minutes)
- Summarize the key points discussed: definitions of speed and velocity, the formula for velocity, and how to apply it in real-life situations.
- Conduct a brief interactive quiz or poll to reinforce the concepts.
- Prepare students for the next session by introducing a topic related to acceleration and how it relates to velocity.

Extended Activities
- Create a project where students track their own mode of transport for a week (walking, biking, etc.) and calculate their velocities. They can present their findings with graphs.
- Assign a research task for students to find and report on how different professions utilize speed and velocity (e.g., sports coaches, pilots, etc.).

Teacher Self-Evaluation:

WEEK 7: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Money

Specific Learning Outcomes:
- By the end of the lesson, the learner should be able to:
- Identify the formula for calculating acceleration in real life situations.
- Work out acceleration in real life situations.
- Acknowledge the application of acceleration in real life situations.

Key Inquiry Question(s):
- How do we calculate acceleration in real life situations?

Learning Resources:
- Top Scholar Maths pg 129-130
- Calculators
- Teacher's Guide
- Charts
- Digital device
- Video clip

Organisation of Learning:

Introduction (5 minutes)
- Begin with a quick review of the previous lesson to activate prior knowledge.
- Guide learners in reading and discussing relevant content from the learning resources, focusing on the essential concepts related to acceleration.

Lesson Development (30 minutes)

Step 1: Discuss the Meaning of Acceleration
- In groups, have learners discuss what acceleration means. Provide examples from everyday life (e.g., driving a car, riding a bicycle).
- Encourage each group to share one example with the class.

Step 2: Identify the Formula for Acceleration
- Guide the students to identify the formula for finding acceleration (a = (final velocity - initial velocity) / time).

- Ensure each group writes down the formula, and discusses how it applies to their examples.

Step 3: Explore Digital Resources
- Allow groups to use digital devices to find video clips or animations that illustrate how to calculate acceleration.
- As they explore, they should note down key points and examples shown in the clips.

Step 4: Solve Real-life Problems
- Present different scenarios (like a car speeding up, a runner starting off the blocks) and let learners work out the acceleration using the formula discussed.
- Encourage them to volunteer their answers and explain their thought process.

Conclusion (5 minutes)
- Summarize the key points covered in the lesson, including the formula for acceleration and its real-life applications.
- Conduct an interactive activity, such as a quick quiz or a conceptual game, to reinforce the main topics learned.
- Give a preview of what will be covered in the next session to pique learners' interest.

Extended Activities:
- Acceleration Investigation Project: Have students measure the time taken for different objects (like balls or toy cars) to travel a set distance and calculate their acceleration. They can present their findings to the class.
- Acceleration in Sports: Research and present examples of how acceleration plays a role in different sports, analyzing metrics such as runner's speed, car racing, or cycling.

Teacher Self-Evaluation:

WEEK 9: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Time, Distance, and Speed

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Explain the meaning of longitude.
- Use the globe to identify the longitudes of marked points.
- Enjoy using globes to identify longitudes of marked points.

Key Inquiry Question:
- How do you identify longitudes on a globe?

Learning Resources:
- Top Scholar Maths (pg 131-134)
- Globes
- Digital devices

Organisation of Learning

Introduction (5 minutes):
- Review Previous Lesson: Start by briefly revisiting what the students learned in the last math lesson about measurements.
- Discussion of Key Concepts: Guide learners to read and discuss relevant content from the provided learning resources, focusing on the definition of longitude and its significance in geography.

Lesson Development (30 minutes):

Step 1: Define Longitude (10 minutes)
- Activity: In groups, have students brainstorm and write down their understanding of what longitude means. Encourage them to think about how it is different from latitude.
- Discussion: Have each group share their definitions, then provide a clear and concise explanation of longitude (the imaginary lines that run from the North Pole to the South Pole).

Step 2: Identifying Longitudes on a Globe (10 minutes)
- Activity: Using the globes provided, have students identify the circles that pass through the North and South poles. Explain how these circles represent lines of longitude.

- Demonstration: Show how to find specific longitudes using geographical coordinates on the globe.

Step 3: Marked Points Activity (5 minutes)
- Activity: Provide students with a marked map with various points that have known longitudes. In their groups, they will use the globes to find and identify the longitudes of these marked points.
- Collaborative Sharing: Each group presents the longitudes they found to the class.

Step 4: Fun Fact Sharing (5 minutes)
- Activity: Each group can share one interesting fact they learned about longitude or famous locations that are found along a specific longitude.
- Discussion: Highlight how understanding longitude is not just about mathematics but also about navigation and geography.

Conclusion (5 minutes):
- Summarize Key Points: Review the definitions of longitude and how to identify it using a globe.
- Interactive Activity: Conduct a quick quiz or a game where learners guess the longitude of various world cities based on their positions on the globe.
- Preview Next Topic: Briefly introduce what will be covered in the next lesson (e.g., latitude and how it relates to longitude).

Extended Activities:
- Research Project: Have students research and present on a location's longitude and how that affects its climate/time zone.
- Longitude Adventure: Create an interactive online map activity where students can explore and identify significant locations based on their longitude.
- Create a Longitude Line: In groups, use string to create a model of longitude lines and present their findings about specific locations on these lines.

Teacher Self-Evaluation:

WEEK 9: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Time, Distance and Speed

Specific Learning Outcomes:
- By the end of the lesson, the learner should be able to:
- Identify points that have the same local time.
- Relate longitudes to time on the globe.
- Enjoy using longitudes to determine points with the same local time.

Key Inquiry Question(s):
- How do you identify points with the same local time using longitudes?

Learning Resources:
- Top Scholar Maths (pg 133-134)
- Digital devices (computers, tablets)
- Teacher's Guide
- Globes

Organisation of Learning:

Introduction (5 minutes)
- Begin with a brief review of the previous lesson focused on the concepts of measurements related to time, distance, and speed.
- Guide learners to read and discuss key points from the learning resources, particularly focusing on how time is related to longitude.

Lesson Development (30 minutes)

Step 1: Introduction to Longitudes and Local Time
- Explain the concept of longitudes and how they influence local time. Discuss the significance of the Prime Meridian and how time zones are determined.

Step 2: Group Activity: Research and Data Collection
- In small groups, learners will use digital devices to search the local time and longitude of selected cities around the world. They will fill in a table with their findings, highlighting cities that share the same local time.

Step 3: Identifying Points on the Globe
- Using the drawn globes, each group will locate the cities identified in step 2. They will visually map out where these cities are situated in relation to each other and discuss which ones share the same local time based on their longitudes.

Step 4: Present Findings
- Each group will present their findings to the class, discussing the cities they researched, the local times, and the significance of their longitudes in determining shared times. Encourage peer feedback and questions to foster class interaction.

Conclusion (5 minutes)
- Summarize the key points and learning objectives achieved during the lesson, reiterating the relationship between longitudes and local time.
- Conduct a brief interactive activity, such as a group quiz or a "find-it" game where students identify cities with the same local time.
- Prepare learners for the next session by previewing topics related to global time zones and daylight saving time, posing questions for them to consider.

Extended Activities:
- Activity 1: Create a timeline chart of different countries and their time zones. Students will research how local time varies across the world based on longitude.
- Activity 2: Investigate how technological advancements have impacted communication across different time zones. Write a brief report or create a presentation on findings.
- Activity 3: Explore historical events that occurred simultaneously in different time zones and how they were affected by time differences.

Teacher Self-Evaluation:

WEEK 9: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Time, Distance and Speed

Specific Learning Outcomes
By the end of the lesson, learners should be able to:
- Explain how to determine local time of places on the earth along different longitudes.
- Determine local time of places on the earth along different longitudes.
- Enjoy determining local time of places on the earth along different longitudes.

Key Inquiry Question
- Why does local time vary in different places of the world?

Learning Resources
- Top Scholar Maths pg 135-137
- Digital devices
- Video clips
- Lesson notes
- Illustration charts
- Chalks and Chalkboard
- Teacher's Guide

Organisation of Learning

Introduction (5 minutes)
- Review the previous lesson, asking students to recall any concepts related to time zones and their importance.
- Guide learners to read and discuss relevant content from the learning resources, focusing on the key concepts of longitudes and local time.

Lesson Development (30 minutes)
The lesson development will be split into 4 distinct steps:

Step 1: Discussion on Local Time
- In groups, learners will discuss how local time changes with different longitudes. Prompt them to consider what time zone they live in and how far they are from others (e.g., New York vs. Los Angeles).

Step 2: Watch and Analyze Video Clip
- Use digital devices to search for and watch an illustration clip about determining local time along different longitudes. Students should take notes on important points discussed in the video.

Step 3: Calculate Local Times
- Provide learners with specific longitudes and ask them to calculate the local time for those places based on their understanding of how time differs over longitudes. Encourage collaboration within groups as they work through calculations.

Step 4: Share and Assess
- Each group shares their findings with the class. During this time, provide feedback and clarify any misconceptions regarding time zones and calculations.

Conclusion (5 minutes)
- Summarize key points discussed in the lesson, reiterating how local time varies due to longitudinal differences.
- Conduct a brief interactive activity, such as a quick quiz or a game to identify different time zones around the world.
- Prepare learners for the next session by previewing upcoming topics or posing questions for them to consider, such as, “How do time zones affect global communication?”

Extended Activities
- Research Project: Students can research a country or region that has more than one time zone and prepare a short presentation on how local time is determined there.
- Create a Time Zone Map: Learners can create a colorful time zone map of the world showing different time zones and their relation to the Prime Meridian.
- Explore Daylight Saving Time: Investigate how Daylight Saving Time affects local time and why certain places observe it.

Teacher Self-Evaluation

WEEK 9: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Money

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Identify currencies that are used in different countries.
- Use digital devices and print resources to search for the different currencies used in different countries.
- Acknowledge the currencies used in different countries of the world.

Key Inquiry Questions:
- What is currency?
- Which currencies do you know?

Learning Resources:
- Top Scholar Maths pg 138
- Digital devices (tablets/laptops/smartphones)
- Magazines and newspapers
- Pictures of different currencies

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on measurements and money.
- Ask the students to define currency. Facilitate a brief discussion on their responses.
- Guide learners to read and discuss relevant content from the learning resources, focusing on understanding various currencies.

Lesson Development (30 minutes)

Step 1: Brainstorming (10 minutes)
- In groups of 4-5, learners brainstorm and create a list of currencies they already know, discussing what makes currency special in each context (e.g., design, value, history).

Step 2: Digital Search (10 minutes)
- Each group is assigned digital devices to research currencies used in specific countries. They will compile a list of at least 5 different currencies and their corresponding countries.

- Encourage learners to look up interesting facts about each currency, such as interesting symbols or historical changes.

Step 3: Print Resource Exploration (5 minutes)
- Groups collaborate on using magazines, newspapers, and other print resources to collect images and information about various currencies.
- They will create a collage poster that includes currency images, names of the currency, and a fun fact for each.

Step 4: Presentation and Feedback (5 minutes)
- Each group displays their collage in class for others to view.
- Facilitate a class discussion where groups share what they learned about the different currencies they explored. Provide constructive feedback.

Conclusion (5 minutes)
- Summarize key points: Review what currency is, different types of currencies, and the countries they represent.
- Conduct a brief interactive activity, such as a quick quiz or a currency match-up game to reinforce the material learned.
- Prepare learners for the next session by previewing upcoming topics like exchange rates or the importance of currency in trade.

Extended Activities:
- Country Currency Booklet: Have students create a small booklet where they can showcase 5 different countries, the currency used, and interesting facts or pictures of that currency.
- Currency Exchange Simulation: Students can role-play a market where they trade goods for different currencies based on current exchange rates. They will learn to pause to discuss how varying values affect trading.

Teacher Self-Evaluation:

WEEK 9: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Money

Specific Learning Outcomes:
- By the end of the lesson, the learner should be able to:
1. Explain how to convert foreign currencies to Kenyan currency using the exchange rates.
2. Convert foreign currency to Kenyan currency in real-life situations.
3. Enjoy converting currencies from one form to another in real-life situations.

Key Inquiry Question(s):
- Why do we change currencies from one form to another?

Learning Resources:
- Top Scholar Maths pg 138-140
- Digital devices (tablets, laptops)
- Exchange rates tables
- Teacher's Guide
- Calculators

Organisation of Learning:

Introduction (5 minutes):
- Start the lesson by reviewing the previous session's topic, focusing on understanding measurements.
- Guide learners to read and discuss relevant sections from the learning resources, particularly regarding currency exchange concepts.

Lesson Development (30 minutes):

Step 1: Understanding Exchange Rates (10 minutes)
- Explain what exchange rates are and how they determine the value of one currency against another.
- Ask students to share any experience they have had with foreign currencies.

Step 2: Researching Current Exchange Rates (10 minutes)

- In pairs or small groups, learners use digital devices to look up current exchange rates for popular currencies (e.g., USD, EUR, GBP) against the Kenyan Shilling (KES).
- They should take notes on the exchange rates they find.

Step 3: Converting Currency (5 minutes)
- Using the exchange rates found, guide the learners to perform simple calculations to convert an amount in foreign currency to KES (e.g., converting $100, €50, or £30).
- Provide a few examples on the board for clarity, demonstrating the calculation step-by-step.

Step 4: Sharing Findings (5 minutes)
- Groups will present their findings to the class, discussing the exchange rates they discovered and their calculated conversions.
- Encourage questions and interactions to foster engagement and clarify any doubts.

Conclusion (5 minutes):
- Summarize the key points from the lesson, reinforcing the importance of understanding how to convert currencies.
- Conduct a quick, interactive quiz or activity where students answer questions based on what they learned today.
- Prepare learners for the next session by giving a preview of upcoming topics, focusing on real-life applications of measurements and money.

Extended Activities:
- Currency Conversion Project: Assign students to collect exchange rate information over a week and create a poster presenting their findings. Include calculations for various foreign currencies to KES.
- Role-Playing Activity: Set up a mock marketplace where students can "buy" and "sell" items using different currencies. This will help them actively practice currency conversion in a social context.

Teacher Self-Evaluation:

WEEK 10: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub-Strand: Money

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
1. Explain how to convert Kenyan currency to foreign currency.
2. Convert Kenyan currency to foreign currency using the exchange rates table.
3. Enjoy converting Kenyan currency to foreign currency.

Key Inquiry Question:
How do we convert Kenyan currency to foreign currency?

Learning Resources:
- Top Scholar Maths (pages 141-142)
- Exchange rate tables
- Charts
- Digital devices
- Calculators

Organisation of Learning:

Introduction (5 minutes)
- Begin with a quick review of the previous lesson on measurements.
- Engage learners by guiding them to read and discuss relevant content on currency conversion from the learning resources. Highlight the importance of understanding exchange rates.

Lesson Development (30 minutes)

Step 1: Introduction to Currency Conversion
- Explain the concept of currency conversion and why it is important in our global economy.
- Introduce the exchange rate table, ensuring learners understand what exchange rates represent.

Step 2: Group Discussion
- Divide the class into small groups. Assign each group a specific currency to convert to (e.g., USD, GBP, EUR).
- Have them discuss the different rates they see on the exchange rate table and how to interpret them.

Step 3: Practice Converting Currency
- Using calculators and the exchange rate tables, ask groups to convert a specific amount of Kenyan Shillings (e.g., 1000 KES) to their assigned foreign currency.
- Encourage groups to show their workings and arrive at their answers collaboratively.

Step 4: Sharing Findings
- Have each group share their conversion results with the class.
- Facilitate a class discussion, allowing students to compare their findings and explain their thought processes.

Conclusion (5 minutes)
- Summarize the key concepts covered, such as understanding exchange rates and the process of converting currencies.
- Conduct a brief interactive quiz where students must convert quick amounts using the exchange rates discussed.
- Preview the next session's topic on "Financial Literacy," inviting students to think about real-life applications of currency conversion.

Extended Activities:
1. Have students research and present on the impact of currency fluctuations on international trade.
2. Assign a project where students need to create a travel budget for a trip abroad, using current exchange rates to convert their total expenses into the applicable foreign currency.
3. Organize a classroom market day where students buy and sell fictional goods with different currencies, requiring them to perform conversions as they trade.

Teacher Self-Evaluation:

WEEK 10: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Money

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Explain the meaning of export duty charged on goods.
2. Calculate export duties charged on goods in real-life situations.
3. Develop interest in learning how to calculate export duties charged on goods in real-life situations.

Key Inquiry Question(s):
- What is export duty?
- How do we determine the export duty of different goods in the country?

Learning Resources:
- Top Scholar Maths pg 142-143
- Calculators
- Teacher's Guide
- Digital devices
- Lesson notes

Organisation of Learning:

Introduction (5 minutes)
- Review Previous Lesson: Begin by asking learners about what they learned in the last lesson to refresh their memories.
- Discussion: Guide learners to read relevant content from the learning resources. Focus on understanding what export duty is and its implications on trade.

Lesson Development (30 minutes)

Step 1: Understanding Export Duty (10 minutes)
- Group Discussion: In small groups, learners will research the meaning of export duty using digital devices.
- Sharing Definitions: Each group shares their findings with the class, ensuring everyone understands export duty.

Step 2: Discovering Percentage of Export Duty (10 minutes)
- Investigate Export Duty Rates: Learners will search for information on the percentage of export duty for various goods in Kenya.
- Class Discussion: Come together as a class to compile the different export duty percentages and discuss factors that influence these percentages.

Step 3: Calculating Export Duty (5 minutes)
- Introducing the Formula: Provide learners with a formula for calculating export duty:
- Example Calculation: Work out a sample problem together as a class using the formula.

Step 4: Real-life Situations (5 minutes)
- Group Activity: Each group picks a good and calculates its export duty using a hypothetical value and the percentages discussed. Prepare to share their findings with the class.

Conclusion (5 minutes)
- Summary: Reinforce the key concepts learned: meaning of export duty, how to find its percentage, and the calculation process.
- Interactive Activity: Conduct a quick quiz or fun game where learners match goods to their respective export duties.
- Preview Next Session: Briefly introduce the next topic, perhaps related to import duties or taxation, to pique interest.

Extended Activities:
- Research Task: Assign learners to research about the impact of export duties on local businesses vs. international trade and present their findings in the next class.
- Calculation Challenge: Create a worksheet with different goods and their values. Challenge learners to calculate the export duties based on varying percentages.

Teacher Self-Evaluation:

WEEK 10: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Money

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Define the term "import duty."
- Work out import duties charged on goods in real-life situations.
- Acknowledge the benefits of import duties to a country.

Key Inquiry Question(s):
- How do we calculate import duties charged on goods and services?
- What is import duty?

Learning Resources:
- Top Scholar Maths, pg 143-144
- Calculators
- Digital devices (tablets, laptops, etc.)
- Lesson notes
- Teacher's Guide

Organisation of Learning:

Introduction (5 minutes):
- Review the previous lesson focusing on measurements related to money.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of import duties.

Lesson Development (30 minutes):

Step 1: Define Import Duty
- In groups, learners will use digital devices to search for and define "import duty."
- Encourage each group to note down their findings and share them with the class.

Step 2: Research Import Duty Percentages
- Learners will research online the percentage rates for import duties on various goods and services, particularly focusing on those relevant in Kenya.

- They should also find examples of goods exempt from import duty.

Step 3: Calculation of Import Duty
- Guide the groups to discuss the formula for calculating import duty using the percentages found in Step 2.
- Example Formula: Import Duty = Cost of Goods x Import Duty Percentage.

Step 4: Practical Calculation Exercise
- Each group will be given different scenarios (real-life situations) where they need to calculate the import duty for a set of goods.
- After calculating, each group will share their findings and calculations with the class.

Conclusion (5 minutes):
- Summarize key points covered, including the definition of import duty, percentage rates, and calculation methods.
- Conduct a brief interactive quiz or game to reinforce the main topics discussed (for example, a quick-fire round of questions about the information shared by groups).
- Preview upcoming topics, such as currency exchange and its impact on import duties, and encourage students to think about how import duties can affect consumer prices.

Extended Activities:
- Research Assignment: Ask students to pick a commonly imported good and research its import duty and total cost. They should prepare a short presentation on their findings, discussing how import duties affect market prices.
- Role-Play Activity: Organize a market day simulation where students take on roles of buyers and sellers. They can calculate costs inclusive of import duties to learn how they impact pricing strategies.
- Math Problems Workbook: Provide students with additional word problems related to import duties for extra practice at home.

Teacher Self-Evaluation:

WEEK 10: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub-Strand: Money

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Explain the meaning of the term excise duty.
- Work out excise duty charged on goods and services in real-life situations.
- Acknowledge the goods and services that attract excise duty.

Key Inquiry Question(s):
- What is excise duty?
- How do we calculate the excise duty on goods?

Learning Resources:
- Top Scholar Maths pg 144-145
- Calculators
- Lesson notes
- Digital devices

Organisation of Learning:

Introduction (5 minutes)
- Start with a brief review of the previous lesson to activate prior knowledge.
- Introduce excise duty and its relevance.
- Guide learners to read and discuss relevant content from pages 144-145 of the Top Scholar Maths, focusing on understanding the key concepts.

Lesson Development (30 minutes)

Step 1: Understanding Excise Duty (10 minutes)*
- In groups, learners will use digital devices to research the definition of excise duty. Each group will write down their findings and share with the class to build a common understanding.

Step 2: Identifying Goods and Services (10 minutes)*
- Groups will search for examples of goods that attract excise duty (like tobacco, alcohol, etc.) and document the percentage of excise duty charged on these items.
- Each group will prepare a brief list of examples to share.

Step 3: Calculating Excise Duty (5 minutes)*
- Introduce the formula for calculating excise duty:
- Using examples gathered, guide learners to calculate the excise duty charged on a selected good or service.

Step 4: Sharing Findings (5 minutes)*
- Each group will share their findings, including definitions, examples, and calculated excise duties. Facilitate a discussion around the differences and commonalities in their research.

Conclusion (5 minutes)
- Summarize the key points covered in the lesson: definition of excise duty, types of goods attracting excise duty, and how to calculate it.
- Conduct a brief interactive activity, such as a quiz or a quick game, to reinforce understanding.
- Preview next session's topic, such as different kinds of taxes and their impacts, asking students to consider which taxes they encounter in daily life.

Extended Activities:
- Have students create a brochure or infographic that explains excise duty, including definitions, examples of goods, and calculations.
- Assign a research task where each student picks a good that has excise duty and tracks how the price changes with and without excise duty over a month.

Teacher Self-Evaluation:

WEEK 10: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Money

Specific Learning Outcomes:
By the end of the lesson, students should be able to:
- Explain the meaning of Value-Added Tax (VAT).
- Calculate VAT charged on goods and services.
- Show interest in learning how to calculate VAT on goods and services.

Key Inquiry Questions:
- Why is VAT charged on goods and services?
- What is Value-Added Tax?

Learning Resources:
- Top Scholar Maths (Pages 145-147)
- Calculators
- Teacher's Guide
- Shopping receipts
- Digital devices (tablets/laptops)

Organisation of Learning:

Introduction (5 minutes)
- Begin by briefly reviewing the previous lesson on measurements involving money.
- Ask students to share any prior knowledge they may have about taxes, specifically VAT.
- Introduce the concept of VAT using a few definitions and its importance in the economy.

Lesson Development (30 minutes)

Step 1: Understanding VAT (10 minutes)
- In groups, students use digital devices to research the meaning of VAT.
- Encourage them to define VAT and identify why it's implemented on goods and services.
- Students should discuss their findings within their groups and prepare a brief presentation to share with the class.

Step 2: Analyzing Shopping Receipts (10 minutes)
- Provide each group with shopping receipts.

- Instruct them to identify which items on the receipt are subject to VAT and note the VAT rates.
- Groups will discuss how VAT is reflected in the prices they pay for goods and services.

Step 3: Calculating VAT (5 minutes)
- Guide students to use the formula:
$$ \text{VAT} = \text{Price} \times \text{VAT rate} $$
- In their groups, have them calculate the VAT for specific items on their receipts.
- Share calculations with the class for assessment and feedback.

Step 4: Class Discussion (5 minutes)
- Invite each group to share one good or service they discussed that attracts VAT.
- Facilitate a class discussion on the implications of paying VAT and any questions they may have about the subject.

Conclusion (5 minutes)
- Summarize the key concepts discussed, including the definition of VAT and how to calculate it.
- Conduct a brief interactive quiz using example prices to reinforce the main topics learned.
- Preview the next lesson, such as exploring how government uses tax revenue, and encourage students to think of other instances where they encounter VAT in their daily lives.

Extended Activities:
- Research Assignment: Students can choose a product they frequently buy and research how much VAT contributes to its price. They may present their findings in class in the next session.
- Field Trip: Arrange a visit to a local business to understand how VAT is applied in real-world scenarios.
- Creative Project: Students create a poster explaining VAT, its benefits, and how it works, which can be displayed in class to educate others.

Teacher Self-Evaluation:

WEEK 11: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Approximations and Errors

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify arbitrary units used in estimating measurements.
- Approximate quantities in measurements in different situations.
- Enjoy estimating measurements using arbitrary units.

Key Inquiry Question(s):
- How do we estimate measurements of different quantities?

Learning Resources:
- Top Scholar Maths, pg. 148
- Various arbitrary units (e.g., hand spans, foot lengths, pencil lengths)
- Different objects: Teacher's desk, football field, Math textbook

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on basic measurements and their significance.
- Guide learners in pairs to read and discuss relevant content from the provided resources, focusing on the concept of arbitrary units.

Lesson Development (30 minutes)
Step 1: Understanding Arbitrary Units (10 minutes)
- Discuss what arbitrary units are and provide examples illustrated in the texts.
- Have students brainstorm a list of potential arbitrary units they could use to measure various objects in the classroom.

Step 2: Measuring with Arbitrary Units (10 minutes)
- In pairs, students select several items (e.g., length of the desk, width of the board, height of a stack of books) and use their chosen arbitrary units to measure these quantities.
- Encourage learners to write down their findings and the specific units they used to compare with their peers later.

Step 3: Estimating and Recording Measurements (5 minutes)

- Each group will approximate measurements for larger objects or areas (e.g., the length of the football field using their arbitrary units).
- Students should record their estimations and conclusions regarding their accuracy.

Step 4: Comparison and Reflection (5 minutes)
- Groups will share their findings with the class.
- Facilitate a discussion on varying results and the effectiveness of different arbitrary units for various types of measurements.

Conclusion (5 minutes)
- Summarize the key points discussed, including examples of arbitrary units and methods for estimating measurements.
- Conduct a quick interactive activity (e.g., a game where students estimate object lengths and then measure them) to reinforce the lesson's main ideas.
- Preview upcoming topics related to more advanced measurement techniques or introduction to units of measurement in the next session.

Extended Activities:
- Measurement Scavenger Hunt: Have students perform a scavenger hunt where they use arbitrary units to estimate the size of different items around the school or home.
- Create a Measurement Art Project: Learners can create an art project incorporating different arbitrary measurements and display their findings with explanations of their chosen units.
- Journal Reflection: Encourage students to reflect in a journal about the usefulness of arbitrary units in real life and how they could be applied in everyday situations.

Teacher Self-Evaluation:

WEEK 11: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Approximations and Errors

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Explain how error is calculated using estimations and actual measurements of quantities.
2. Determine errors using estimations and actual measurements of quantities.
3. Appreciate the role of approximations and errors in real-life situations.

Key Inquiry Question(s):
- How do you determine errors in quantities?

Learning Resources:
- Top Scholar Maths, pg 149-150
- Calculators
- Measuring instruments (rulers, tape measures, etc.)
- Teacher's Guide

Organization of Learning:

Introduction (5 minutes)
- Review the previous lesson briefly, focusing on measurement concepts.
- Guide learners to read relevant content from Top Scholar Maths, emphasizing key concepts related to estimation and error.

Lesson Development (30 minutes)

Step 1: Understanding Error Calculation (10 minutes)
- In groups, learners discuss and record how error can be calculated. Introduce the formula for error:
\[\text{Error} = \text{Actual Measurement} - \text{Estimated Measurement} \]
- Encourage groups to share examples of everyday situations where they might encounter measurement errors.

Step 2: Practical Estimation and Measuring (10 minutes)
- Each group selects a few real-world objects (e.g., books, tables) to estimate their measurements (length/width/height) without using tools.

- After estimating, learners then measure the actual dimensions using appropriate instruments.

Step 3: Calculation of Errors (5 minutes)
- Groups calculate the errors for their measurements using the previously discussed formula.
- Each group prepares to share their findings, focusing on both the size of the error and factors that might have influenced their estimates.

Step 4: Peer Sharing (5 minutes)
- Each group presents their findings to the class, highlighting interesting observations and discussing discrepancies between estimates and actual measurements.

Conclusion (5 minutes)
- Summarize the key points of the lesson: how error is calculated, the importance of accurate measurements, and real-life applications.
- Conduct a brief interactive quiz where students answer questions about the day's lesson to reinforce learning.
- Preview the next session's topic related to different types of measuring instruments and their applications.

Extended Activities:
- Real-Life Applications: Have students conduct a home project where they measure different objects, compare their estimates with actual measurements, and report their findings in class.
- Error Analysis Worksheet: Provide students with a worksheet that includes different scenarios where they need to estimate and calculate errors in measurements given.
- Group Presentation: Assign groups to research different fields that rely on precise measurements (like construction, science laboratories, etc.) and present how they handle estimation and error.

Teacher Self-Evaluation:

WEEK 11: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Approximations and Errors

Specific Learning Outcomes:
By the end of the lesson, students should be able to:
1. Identify the formula for calculating percentage error.
2. Determine percentage errors using actual measurements of quantities.
3. Enjoy working out percentage errors of quantities.

Key Inquiry Question(s):
- How do you calculate the percentage error of quantities?

Learning Resources:
- Textbook: Top Scholar Maths (pg 150-151)
- Digital devices (tablets or computers)
- Calculators
- Video clip on percentage error
- Teacher's Guide

Organisation of Learning:

Introduction (5 minutes)
- Review the content from the previous lesson about measurements.
- Ask students to recall any challenging areas they faced in their last exercise.
- Introduce the topic of percentage error by discussing its importance in real-life situations, such as engineering and science.
- Guide learners to read and discuss the relevant content from the textbook, focusing on the key concepts of percentage error.

Lesson Development (30 minutes)
Step 1: Watch and Learn (10 minutes)
- Divide the class into small groups.
- Instruct each group to use digital devices to search for and watch a video clip that explains how to calculate percentage error.
- After the video, ask each group to summarize the main points they learned.

Step 2: Identify the Formula (5 minutes)

- Ask the groups to identify and write down the formula for calculating percentage error from the video.
- Have each group share the formula they found:
$$ \text{Percentage Error} = \left(\frac{\text{Estimated Value} - \text{Actual Value}}{\text{Actual Value}} \right) \times 100 $$
- Facilitate a brief discussion to ensure everyone understands each part of the formula.

Step 3: Calculate Percentage Errors (10 minutes)
- Provide students with a set of measurements (both estimated and actual).
- Instruct groups to work together to calculate the percentage error using the formula.
- Circulate around the classroom to support students as they work through the calculations and provide feedback.

Step 4: Share Findings (5 minutes)
- Ask each group to present their calculations and findings to the class.
- Encourage other students to ask questions or give feedback on the different methods used to calculate percentage error.

Conclusion (5 minutes)
- Summarize key points covered in the lesson: the definition of percentage error, the formula used to calculate it, and examples worked on in class.
- Conduct a brief interactive activity where students can call out examples of where they might use percentage error in daily life, reinforcing the concepts discussed.
- Preview the next session, hinting at how percentage error can affect data interpretation and decision-making.

Extended Activities:
- Assign students a research project where they find real-world examples of percentage error in various fields like science, finance, or technology and present their findings to the class.
- Encourage students to create a PowerPoint presentation or poster about percentage error and its applications, which they can present in a future lesson.

Teacher Self-Evaluation:

WEEK 11: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Measurements
Sub Strand: Mixed Exercise 3

Specific Learning Outcomes:
- By the end of the lesson, the learner should be able to work out questions on Mixed Exercise 3.
- Embrace teamwork as they work through the problems in Mixed Exercise 3.

Key Inquiry Question(s):
- How can collaborating with peers enhance our understanding of measurement problems?

Learning Resources:
- Top Scholar Maths, pp. 152-153
- Calculators
- Teacher's Guide

Organisation of Learning:

Introduction (5 minutes)
- Begin by reviewing the key concepts covered in the previous lesson about measurements.
- Ask students to recall any measurement formulas or concepts they found challenging.
- Introduce Mixed Exercise 3 and explain its importance in developing their measurement skills.

Lesson Development (30 minutes)
- **Step 1:** Group Formation
- Divide the class into small groups of 3-4 students.
- Ensure each group has access to the Top Scholar Maths text and calculators.

- **Step 2**: Problem Distribution
- Assign each group a specific set of problems from Mixed Exercise 3.
- Encourage students to discuss and list any relevant formulas they may need to solve their assigned problems.

- Step 3: Collaboration and Problem Solving
- In their groups, students will collaborate to solve the assigned problems, ensuring that each member participates.

- Circulate the classroom to assist groups as needed, providing hints or guidance without giving away answers.

- **Step 4:** Group Presentation
- After solving the problems, each group will present one problem to the class, explaining their thought process and solution.
- Encourage questions from peers to foster discussion and deeper understanding.

Conclusion (5 minutes)
- Summarize the key points learned during the lesson, highlighting the importance of teamwork in solving mathematical problems.
- Conduct a brief interactive activity, such as a quick quiz where students raise their hands for true/false statements related to what they've learned.
- Preview the next session's topics, such as advanced measurement concepts or real-world applications of measurements, and pose questions for them to think about at home.

Extended Activities:
- Create a measurement scavenger hunt where students must measure various objects around the school using different measurement units (e.g., meters, liters, grams).
- Ask students to create their own measurement problems based on real-life scenarios and exchange them with other students for solving.
- Encourage students to explore online resources or videos that explain measurement concepts and invite them to share interesting findings with the class.

Teacher Self-Evaluation:

WEEK 11: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Geometry
Sub Strand: Coordinates and Graphs

Specific Learning Outcomes:
- By the end of the lesson, the learner should be able to:
- Explain the steps to follow when plotting points on a Cartesian plane.
- Plot points accurately on a Cartesian plane.
- Enjoy the activity of plotting points on a Cartesian plane.

Key Inquiry Question:
- How do we plot points on a Cartesian plane?

Learning Resources:
- Top Scholar Maths textbook
- Graph paper
- Pencils
- Rulers

Organisation of Learning:

Introduction (5 minutes)
- Begin with a quick review of the previous lesson, discussing basic concepts of coordinates and their importance in geometry.
- Ask students to read and discuss relevant sections from the Top Scholar Maths textbook, focusing on the Cartesian plane and coordinate systems.

Lesson Development (30 minutes)

Step 1: Understanding the Cartesian Plane (10 minutes)
- Introduce the concept of the Cartesian plane, explaining the x-axis and y-axis.
- Show how the axes intersect at the origin (0,0) and how coordinates are represented as (x, y).

Step 2: Discussing the Steps to Plot Points (5 minutes)
- Facilitate a discussion where students articulate the steps for plotting points on the Cartesian plane:
1. Identify the x-coordinate.
2. Move horizontally to the right (if positive) or left (if negative).

3. Identify the y-coordinate.
4. Move vertically up (if positive) or down (if negative).
5. Mark the point where the lines intersect.

Step 3: Hands-on Practice (10 minutes)
- In pairs or groups, have learners draw Cartesian planes on their graph paper.
- Provide a list of coordinates (e.g., (2,3), (-1,-2), (0,4)) for students to plot.
- Instruct learners to write down the coordinates next to the plotted points.

Step 4: Sharing and Feedback (5 minutes)
- Invite groups to share their plotted points with the class.
- Encourage students to give constructive feedback and answer any questions.

Conclusion (5 minutes)
- Summarize the key points learned regarding how to correctly plot points on the Cartesian plane.
- Conduct a brief interactive activity, such as a "Coordinate Guessing Game," where students must guess coordinates based on provided clues.
- Preview the next lesson on graphing linear equations, hinting at how plotting points will relate to that topic.

Extended Activities:
- Coordinate Art: Encourage students to create art by plotting points and connecting them to form shapes (e.g., triangles, squares).
- Digital Tools: Introduce online graph plotting tools, allowing students to experiment with plotting and understanding more complex coordinates.
- Coordinate Scavenger Hunt: Create a scavenger hunt where students find physical locations on the school grounds based on given coordinate pairs.

Teacher Self-Evaluation:

WEEK 12: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Geometry
Sub Strand: Coordinates and Graphs

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Explain how to draw a straight line graph given an equation.
- Draw a straight line graph given an equation.
- Enjoy generating values and drawing straight lines graph.

Key Inquiry Question(s):
- How do you draw a straight line graph given an equation?

Learning Resources:
- Top Scholar Maths pg 155-156
- Graph books
- Pencils
- Rulers

Organisation of Learning:

Introduction (5 minutes)
- Begin the lesson by reviewing the previous session's content on the Cartesian plane.
- Encourage students to share what they remember about plotting points and understanding equations.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing understanding the key concepts of linear equations.

Lesson Development (30 minutes)

Step 1: Understanding Equations of Straight Lines (10 minutes)
- Introduce the general form of a linear equation (e.g., $y = mx + b$).
- Discuss how to identify the slope (m) and y-intercept (b) from the equation and their implications on the graph.
- Ask students to interpret several examples and predict the shape of the graph based on different values of m and b.

Step 2: Generating Values (10 minutes)

- In groups, students will select or be given a linear equation.
- Their task is to generate a table of values (for example, choosing x-values and calculating the corresponding y-values).
- Remind them to choose a range of x-values to see how the line behaves.

Step 3: Plotting the Graph (5 minutes)
- Students will take their tables and plot the points on the Cartesian plane using graph books and rulers.
- Ensure they know to plot each point carefully and remind them to label the axes.

Step 4: Joining Points and Sharing Work (5 minutes)
- Once the points are plotted, guide students to draw straight lines through these points.
- Allow time for groups to share their graphs with the class. Encourage peer assessment and constructive feedback.

Conclusion (5 minutes)
- Summarize key points learned, such as the importance of recognizing the slope and intercept, as well as generating values and plotting them accurately.
- Conduct a brief interactive quiz or game to reinforce the main topics discussed during the lesson.
- Preview the next session on different types of linear equations or introducing curves, encouraging students to think about how these concepts might expand.

Extended Activities:
- Encourage students to create their own linear equations and exchange them with a partner to generate tables, plot, and graph each other's equations.
- Suggest an online graphing tool or application that students can use to visualize transformations of linear equations by changing the slope and y-intercept.
- Assign a small project where students find real-world examples of linear relationships and create graphs based on their findings.

Teacher Self-Evaluation:

WEEK 12: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Geometry
Sub Strand: Coordinates and Graphs

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Outline the steps for drawing parallel lines on a Cartesian plane.
- Draw parallel lines on the Cartesian plane.
- Enjoy drawing parallel lines on a Cartesian plane given equations.

Key Inquiry Question(s):
- How do we draw parallel lines on a Cartesian plane?

Learning Resources:
- Graph books
- Top Scholar Maths pg 157-158
- Rulers
- Pencils

Organisation of Learning

Introduction (5 minutes):
- Review the previous lesson on linear equations and their graphs.
- Ask students to share their understanding of parallel lines and their characteristics.
- Guide learners to read and discuss relevant content from pages 157-158 of Top Scholar Maths, focusing on how parallel lines are represented in equations.

Lesson Development (30 minutes):

Step 1: Discuss Characteristics of Parallel Lines
- Explain that parallel lines have the same slope but different y-intercepts.
- Introduce the equations of parallel lines (e.g., $y = 2x + 1$ and $y = 2x - 3$).

Step 2: Generate Tables of Values
- In groups, have students select two pairs of parallel line equations.
- Each group generates a table of values for both equations (x-values from -3 to 3).

Step 3: Plot the Lines on Cartesian Plane

- Using their tables of values, students plot the points for both equations on graph paper.
- Once points are plotted, instruct them to join the points to form straight lines.

Step 4: Measure the Distance Between Lines
- Provide each group with a set square.
- Students should choose two points on each of their lines and use the set square to measure the perpendicular distance between the two parallel lines at those points.

Conclusion (5 minutes):
- Summarize the key points discussed: the characteristics of parallel lines, how to find their equations, and the steps to draw them.
- Conduct a quick interactive quiz, asking students to identify key characteristics of parallel lines.
- Preview the next lesson, which will cover perpendicular lines and their equations, encouraging students to think about the differences between parallel and perpendicular lines.

Extended Activities:
- Homework Assignment: Have students find real-life examples of parallel lines (e.g., roads, railroad tracks) and sketch or take photos to present in the next class.
- Challenge Activity: Encourage students to create their own equations for parallel lines and exchange them with a peer to draw and measure the distance.

Teacher Self-Evaluation:

WEEK 12: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Geometry
Sub-Strand: Coordinates and Graphs

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Define the term parallel lines.
- Relate the gradients of parallel lines.
- Acknowledge the relationship of the gradients to the parallel lines.

Key Inquiry Question(s):
- What are parallel lines?

Learning Resources:
- Top Scholar Maths pp. 159-161
- Rulers
- Graphing Pencils

Organisation of Learning

Introduction (5 minutes)
- Begin with a brief review of the previous lesson focusing on basic concepts of lines and slopes.
- Ask students to read relevant sections on parallel lines from "Top Scholar Maths" pp. 159-161 and discuss their understanding in pairs, emphasizing key concepts about parallel lines.

Lesson Development (30 minutes)

Step 1: Definition of Parallel Lines (7 minutes)
- Discuss and define parallel lines in the context of geometry.
- Explain that parallel lines are lines in a plane that never meet and are always the same distance apart.
- Use visual examples to demonstrate this concept.

Step 2: Understanding Gradients (7 minutes)
- Introduce the concept of gradient (slope) and how to calculate it using the formula:
- Provide examples of two parallel lines and calculate their gradients step-by-step.

Step 3: Group Activity – Generating Table of Values (10 minutes)
- Divide the class into small groups.
- Assign each group different linear equations that represent parallel lines (e.g., $y = 2x + 3$ and $y = 2x - 5$).
- Instruct them to create a table of values for each line and plot the points on graph paper. Use rulers to ensure accuracy.

Step 4: Comparing Gradients (6 minutes)
- Have each group calculate and write down the gradients of their assigned lines.
- Prompt a class discussion comparing the gradients found by each group. Reinforce the idea that parallel lines have identical gradients.

Conclusion (5 minutes)
- Summarize key points regarding the definitions and characteristics of parallel lines and their gradients.
- Conduct an interactive quiz or quick discussion to reinforce learning (e.g., "What happens if two lines have the same gradient?").
- Preview the next session on intersecting lines and how their gradients differ from those of parallel lines.

Extended Activities:
- Independent Practice: Assign worksheet problems where students can practice finding the gradients of various equations, including those of parallel lines.
- Real-Life Connections: Have students find examples of parallel lines in their environment and explain how they might calculate their slopes graphically or algebraically.
- Project: Encourage students to create a poster illustrating parallel lines and their properties, complete with examples and graphs.

Teacher Self-Evaluation:

WEEK 12: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Strand: Geometry
Sub Strand: Coordinates and Graphs

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Outline the steps for drawing perpendicular lines on the Cartesian plane.
- Draw perpendicular lines on the Cartesian plane.
- Enjoy drawing perpendicular lines on the Cartesian plane.

Key Inquiry Question:
- How do we know lines are perpendicular on a Cartesian plane?

Learning Resources:
- Top Scholar Maths, pg 161-163
- Rulers
- Graph books
- Pencils

Organisation of Learning:

Introduction (5 minutes)
1. Begin with a quick review of the previous lesson on lines and their properties.
2. Ask students guiding questions to stimulate discussion. For example: “What do we remember about perpendicular lines?”
3. Guide learners to read and discuss relevant content from the learning resources (pages 161-163), emphasizing the definition and visual representation of perpendicular lines.

Lesson Development (30 minutes)

Step 1: Understanding Perpendicular Lines (10 minutes)
- Discuss the definition of perpendicular lines and their properties. Explain that two lines are perpendicular if they intersect at a right angle (90 degrees).
- Show examples on the board and highlight their graphical representation on a Cartesian plane.

Step 2: Outlining Steps (10 minutes)
- In groups, have students outline the steps to draw perpendicular lines. Prompt them with questions:

- What do we need to start?
- How can we find the correct angle for perpendicularity?
- Ask groups to share their steps with the class, and compile the information on the board.

Step 3: Generating Table of Values (5 minutes)
- Assign different equations of lines (e.g., $y = mx + b$, where m is the slope) that are known to be perpendicular.
- Have students generate a table of values for given perpendicular line equations.

Step 4: Plotting and Measuring (5 minutes)
- Guide students to plot their lines on graph paper using their tables of values.
- After plotting, have each group measure the angle at the intersection point of their two lines using a protractor. Discuss whether the lines are indeed perpendicular based on their measurements.

Conclusion (5 minutes)
1. Summarize the key points learned during the lesson about perpendicular lines and their characteristics.
2. Conduct a quick interactive quiz or game where students identify whether pairs of lines are perpendicular based on their slopes or graphical representations.
3. Preview the next session by introducing the topic of parallel lines and their properties, encouraging students to think about how they relate to perpendicular lines and what different properties they have.

Extended Activities:
- Challenge Homework: Ask students to find examples of perpendicular lines in real life and bring drawings or photographs to the next class.
- Graphing Software: Have learners use graphing software or online graphing tools to draw and manipulate different pairs of lines, observing their properties dynamically.
- Create a Game: Students can create a board game where they have to plot lines and determine if they are perpendicular as part of the gameplay.

Teacher Self-Evaluation:

WEEK 12: LESSON 5

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	MATHEMATICS			

Subject: Geometry
Strand: Coordinates and Graphs

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Relate the gradients of perpendicular lines.
- Apply graphs of straight lines in real-life situations.
- Acknowledge the application of straight lines in real life.

Key Inquiry Question(s):
- How do we establish the relationship of perpendicular lines?

Learning Resources:
- Top Scholar Maths pg 163-167
- Graph books
- Pencils
- Rulers

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on linear equations and their graphs.
- Engage learners in a discussion about what they remember, guiding them to read relevant content from the learning resources that covers gradients and straight lines. Emphasize the significance of gradients in determining the nature of lines, particularly perpendicular lines.

Lesson Development (30 minutes)

Step 1: Generating a Table of Values (10 minutes)
- Activity: In small groups, learners will receive two different linear equations (for example, $y = 2x + 1$ and $y = -0.5x + 3$). Each group will generate a table of values for their equations.
- Guide Questions:
- What x-values will you choose to create your table?
- How do you calculate the corresponding y-values for your selected x-values?

Step 2: Drawing the Lines (10 minutes)

- Activity: Once tables are completed, learners will plot their points on graph paper and draw the respective straight lines.
- Guide Questions:
- Are the lines intersecting or parallel?
- How can you tell if your lines are perpendicular just by looking at their slopes?

Step 3: Finding Gradients (5 minutes)
- Activity: Learners will calculate the gradients (slopes) of their lines using the formula (change in y)/(change in x).
- Discussion: Groups will compare their gradients and discuss what this tells them about the relationship between the lines.
- What do you notice about the products of the slopes of perpendicular lines?

Step 4: Connecting to Real Life (5 minutes)
- Activity: Discuss real-life situations where straight line graphs are applicable (e.g. cost versus quantity, distance over time).
- Guide Questions:
- Can you think of any examples where these equations might model a real scenario?
- How can we use graphs to make predictions in these scenarios?

Conclusion (5 minutes)
- Summarize the key points discussed in the lesson, particularly the relationships between gradients of perpendicular lines and applications of straight lines.
- Conduct a brief interactive quiz: Ask learners to identify whether given situations describe perpendicular, parallel, or intersecting lines.
- Preview the next lesson's topic: Introduction to systems of equations and how they can be visually represented on graphs.

Extended Activities:
- Activity: Have students research a profession (like architecture or economics) where understanding gradients and linear equations is crucial. They can create a presentation on how these concepts are applied in their chosen field.
- Graphing Challenge: Provide learners with real-world data (like household expenses or sports statistics) and have them create their own linear models through tables and graphs.

Teacher Self-Evaluation: